

Dynamical spectral functions from bitstring-sampled quantum subspaces: entanglement, not one-body magic, tracks the sampling cost

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September 5, 2026

Abstract

Sample-based quantum diagonalization (SQD) and quantum-selected configuration interaction (QSCI) are the electronic-structure methods with the most quantum-hardware traction, yet their canonical target—the ground-state *energy*—is precisely where classical methods have caught up: matchgate simulators, autoregressive neural samplers, and heat-bath configuration interaction now match or beat the quantum-sampled subspace. We move the target to *dynamics* and to the underlying *resource* question. From one bitstring-sampling primitive—computational-basis measurements of a shallow real-time circuit, with no Hadamard or controlled-unitary tests—we reconstruct, from sampled subspaces, the single-particle spectral function $A(\omega)$ and its momentum-resolved form $A(k, \omega)$; the same primitive targets the neutral-sector dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$, each channel assembled classically in the Lehmann representation from its own sector-specific subspace. The sampled reconstruction matches exact diagonalization on Hubbard chains and, for the orbital-projected $A(\omega)$, across a nineteen-molecule suite (energies FCI-verified to $< 10^{-5}$ Ha), and runs on the IBM Heron processor.

Second, we ask which resource controls the cost—the determinant support $|\mathcal{S}|$ the sampler must populate—and correct a natural expectation. On the number-conserving states of chemistry the fermionic AntiFlatness (fermionic magic) collapses to a single one-particle-reduced-density-matrix invariant, $\mathcal{F}_1 = 4 \operatorname{tr}[\gamma(1 - \gamma)]$ (equal to $2N_u$ for spin-restricted natural orbitals), the effectively-unpaired-electron index and one-body-entanglement linear entropy. Being an orbital-rotation (Gaussian) invariant while $|\mathcal{S}|$ is basis dependent, $\mathcal{F}_1$ is provably *decoupled* from the cost, two-sidedly and without a stable sign. The cost is instead *lower-bounded and empirically tracked* by the basis-dependent entanglement—the minimal bond dimension χ . On the molecular suite the two candidate resources are not distinguishable (Spearman $\rho = 0.90$ for χ versus 0.86 for $\mathcal{F}_1$, $n = 38$, where both merely track static correlation); what separates them is the Hubbard interaction sweep, where χ holds at $\rho = 0.57$ while $\mathcal{F}_1$ falls to $\rho = -0.11$ ($n = 30$, not significant). One-body fermionic magic is thus a faithful multireference diagnostic but an unreliable predictor of the sampling cost; any genuine advantage lives one stratum above, in the high-rank non-Gaussianity of the connected higher-body cumulants in the operative sampling frame. We prove moment exactness and a sampling-capture bound polynomial in $|\mathcal{S}|$ and independent of the Hilbert-space dimension. On the artificial-intelligence side of the hybrid loop, self-consistent configuration recovery improves the sampled subspace under realistic device noise, while in the dequantization test the large effect is classical and free: on an under-sampled device head an Epstein–Nesbet perturbative tail, drawing no additional shots, cuts the energy error by 31% ($39.24 \pm 3.95 \rightarrow 26.98 \pm 0.66$ mHa over five seeds, 3.1 s.d. of the pre-correction spread), whereas a *learned* generative model adds a further 4.0 mHa at only 1.2 s.d. of its own spread and does not beat that baseline—delimiting the quantum–AI frontier.

1. Introduction

The promise that quantum computers will solve the central problem of electronic structure—computing the ground-state energy of a many-electron Hamiltonian, whose configuration space grows

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combinatorially and whose exact solution is QMA-complete in the worst case [62]—has driven a decade of algorithmic development [4, 5, 111]. The variational quantum eigensolver (VQE) [1, 2] dominated the noisy-device era but has, for the purpose of demonstrating a scaling advantage, largely run its course: barren plateaus render deep expressive circuits untrainable [3], and the very structural conditions that let a circuit avoid them tend also to render it classically simulable [6]. Sample-based quantum diagonalization (SQD) [7], equivalently quantum-selected configuration interaction (QSCI) [8], took a different route: rather than optimize a circuit to *be* the ground state, one uses a shallow, noise-tolerant circuit only to *sample* the electronic configurations that carry appreciable weight, and then diagonalizes the Hamiltonian classically in the sampled determinant subspace on high-performance hardware [7, 12–14]. Within two years SQD became the de facto default of pre-fault-tolerant quantum chemistry, reaching iron–sulfur clusters at 77 qubits with the Fugaku supercomputer in the loop [7, 15] and extending to excited states [11], adaptive state preparation [9], density-matrix embedding [17], and auxiliary-field quantum Monte Carlo [16, 109, 128].

Yet the target of SQD—the ground-state *energy*—is exactly where the classical frontier has closed the gap. The broader case for a generic exponential quantum advantage in ground-state chemistry was already found wanting: across sampled chemical space, heuristic classical methods remain competitive and the assumed advantage rests on efficient state preparation [20]. For SQD specifically, an extended matchgate simulator reproduces the unitary-cluster-Jastrow-SQD workflow [21], single-layer unitary cluster Jastrow energies are computable in polynomial time—reproducing the flagship 77-qubit experiment on a laptop at *lower* energy [22]—and classical autoregressive neural samplers reconstruct the determinant-selection loop on a GPU [23, 24]. A direct quantum-chemistry benchmark finds classical heat-bath configuration interaction more compact and cheaper than QSCI at accessible sizes [25], and a complementary lattice-model study reports the intrinsic exponential configuration-space scaling that drives this [26]—a conclusion we reached independently in a companion review [27]. This mirrors the fate of the first “quantum utility” demonstration [28], reproduced classically within months by tensor-network [29] and sparse-Pauli [30] methods. The lesson is quantitative: under fault-tolerant overheads even polynomial speedups do not deliver practical advantage [31], so a *defensible* advantage must live where the classical cost genuinely explodes [32].

We argue that the *defensible* quantum advantage of a sample-based method, if one exists, is not in the energy but in the *sampling* of a hard distribution, realized most cleanly for *dynamical* observables—the regime our primitive targets, a necessary precondition for any such advantage rather than a demonstration of one. Real-time dynamics is where classical tensor networks hit the entanglement barrier [61], where the fermion sign problem becomes a phase problem [60], and where every surviving beyond-classical hardware demonstration now lives—from the random-circuit-sampling programme [34] and verifiable-advantage correlators [33] to real-time simulations of the Fermi–Hubbard model that already outrun exact classical methods on superconducting [35] and trapped-ion [36] processors, and to precision estimation of prethermal dynamics beyond classical reach [37, 38]. The natural dynamical targets are the *spectral functions* of correlated matter: the single-particle spectral function $A(\omega)$ measured by angle-resolved photoemission [102], its momentum-resolved form $A(k, \omega)$, and the dynamical structure factor $S(q, \omega)$ probed by neutron and X-ray scattering [58, 59]—the central outputs of dynamical mean-field theory [101]. Classically these are obtained by dynamical density-matrix renormalization [89, 91], time-dependent DMRG [92], and kernel-polynomial expansions [93] in one dimension, where the density-matrix renormalization group is near-optimal [87, 131], but they remain hard beyond it.

Crucially, no published method reconstructs a frequency-resolved single-particle spectral function from a *bitstring-sampled* quantum subspace in the charged ($N \pm 1$) sectors, with the Green’s function assembled classically in the Lehmann representation. The SQD/SKQD/TE-QSCI lineage stops at energies and, at most, a handful of discrete excited-state levels [10, 11]; quantum Green’s-function methods use Hadamard tests [47, 48] or Krylov and quantum-subspace recursions built from measured overlaps [41–43, 45, 46, 130]—the last computing a dynamical structure factor $S(q, \omega)$ for a Kitaev spin liquid from a measured-overlap subspace rather than sampled configurations—or hybrid quantum–classical response-and-structure frameworks [52], rather than sampled subspaces;

and the hardware real-time-observable results come from a separate error-mitigation-plus-Trotter line [37–39]. The two closest results—a QSCI quasiparticle band structure that returns discrete poles [18] and a molecular spectrum obtained from a neutral-sector autocorrelation built from short-time quantum-evolved samples with classically projected long-time dynamics [51]—return static levels or a neutral-sector response, not the charged-sector ($N\pm 1$) single-particle Green’s function that our Lehmann construction assembles from the sampled subspace. We occupy that gap.

To know *where* such a method is hard—and, just as important, where it is not—one must know which resource controls its cost. Free-fermion (matchgate) circuits are classically efficient [64–67]; the resource that leaves this manifold is fermionic non-Gaussianity, and every pure non-Gaussian fermionic state is a *magic* state for matchgate computation [68]—the fermionic counterpart of the non-stabilizerness that powers universal qubit computation [69, 70]. The efficiently computable fermionic AntiFlatness (FAF) [71], built from the Majorana covariance matrix, quantifies that distance from the Gaussian manifold, and it is natural to hope that this single measured number decides when the quantum sampler is essential. *It does not*. The cost of a sample-based method is its determinant-subspace size $|\mathcal{S}|$ (defined below); we prove one-body fermionic magic is decoupled from it. Understanding this decoupling—and locating the genuine hardness in the higher-order sector, where the paired “magic-input” regime is classically tractable for additive-error *estimation* under free-fermionic dynamics [77], though computational-basis *sampling* of the same family in a rotated frame may remain hard [129]—is the second contribution of this work.

This work makes *two* central contributions to quantum computation, each closing a gap that a comprehensive survey of the literature confirms is open, and each supported by a rigorous account of its scope.

First, a new use of the sampled-subspace toolkit: frequency-resolved dynamical response from bitstring-sampled subspaces—charged-sector single-particle spectra and neutral-sector structure factors. We compute the single-particle spectral function $A(\omega)$ and, from the same charged- $(N\pm 1)$ sampled configurations, its momentum-resolved form $A(k, \omega)$; the density and spin responses $S(q, \omega)$ and $S^{zz}(q, \omega)$ follow from number-conserving seeds in the neutral N sector—all purely from computational-basis bitstrings, with no Hadamard tests, no controlled unitaries, no overlap estimation. This lifts the entire SQD/QSCI toolkit from the static energies and discrete levels it has been confined to [7, 8, 11, 18] to continuous, frequency-resolved dynamics, and we validate it against exact diagonalization on Hubbard chains, across nineteen molecules spanning every bonding regime, and on the IBM Heron processor.

Second, the resource that sets the cost—and the one that does not. The cost of a sample-based method is the size $|\mathcal{S}|$ of the determinant subspace the sampler must populate to reach a target accuracy, and we identify what controls it by first ruling out what does not. Restricted to the number-conserving states of chemistry, the fermionic AntiFlatness [71] collapses to a single one-particle reduced-density-matrix invariant, $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ (equal to $2N_u$ for spin-restricted natural orbitals)—the effectively-unpaired-electron index [112, 113] and the one-body-entanglement linear entropy [115] (§3). Being a function of the orbital-rotation-invariant occupation spectrum it is a Gaussian invariant, whereas $|\mathcal{S}|$ is basis dependent; a quantity unchanged by the orbital rotations that sweep $|\mathcal{S}|$ over its whole range cannot predict it. Across the same nineteen FCI-verified molecules one-body magic does track static, multireference character—switching on for covalent bond-breaking, already large for inherently multireference C_2 and O_2 , near-Gaussian for the ionic bond of LiF , refining to fermionic non-Gaussianity the qubit non-stabilizerness peak of molecular bonding [81]—yet this correlation is not even sign stable: a free-fermion determinant has $\mathcal{F}_1 = 0$ with exponentially large $|\mathcal{S}|$, and increasing on-site repulsion raises $\mathcal{F}_1$ while *lowering* $|\mathcal{S}|$. One-body fermionic magic is a faithful multireference diagnostic but an unreliable predictor of the sampling cost, which is set instead by the basis-dependent entanglement—the minimal bond dimension χ , which tracks $|\mathcal{S}|$ across the suite. Any genuine quantum advantage lives one stratum above, in the generic, high-rank non-Gaussianity of the connected higher-body cumulants in the fixed sampling basis—not the orbital-invariant higher-order antiflatness $\mathcal{F}_{k\geq 2}$, which we exhibit large on a classically trivial paired family—beyond the stratum whose free-fermionic estimation primitives remain classically

tractable [77]. To our knowledge this is the first time a fermionic-magic measure is connected to sample-based diagonalization.

These contributions are delivered with the rigor the dequantization debate now demands [20, 22, 25]: we scope the reach of the method (it is asymptotic, and lives in two-dimensional and chemical settings, not in one dimension where tensor networks remain near-optimal [87, 131]), prove a moment-exactness and polynomial-shot bound for the sampled reconstruction, and, on the artificial-intelligence side of the hybrid loop, show that self-consistent configuration recovery [7] improves the sampled subspace under realistic device noise [63, 119]—through valid-shot economy rather than quantum advantage—while a *learned* generative tail-completer does *not* beat that classical baseline (Fig. 18), delimiting the quantum–AI frontier consistent with, not counter to, the classical-simulability frontier we and others have charted [25, 27]. The resulting resource diagnostics, computed from bitstrings already in hand, map where that frontier lies for the spectral and dynamical workloads of quantum-centric computation [7, 15].

Beyond the specific algorithm, the reframing carries weight for both the science and the industry of quantum computing. *For the field*, it forges a first link between the resource theory of fermionic non-Gaussianity [68, 71] and sample-based electronic-structure computation [7, 8], and replaces a natural but mistaken hope—that a one-body magic number certifies where sampling is hard—with a precise statement of which resource sets the cost and which does not. *For practice*, the observables it delivers—the single-particle spectral function, its momentum resolution, and the dynamical structure factor of correlated matter [101, 102]—are precisely those that materials discovery, catalysis, and correlated-electron device design demand and that the energy-only output of today’s sample-based hardware does not provide; and because the resource diagnostics are measured from bitstrings already in hand, they chart, before further runtime is spent, where the classical frontier lies for a given spectral or dynamical workload on the quantum-centric supercomputers now in operation. We develop both threads in Sec. 6.

2. The method: spectral functions from bitstring-sampled subspaces

2.1 Construction

For a Hamiltonian $\hat{H}$ with N -electron ground state $|0\rangle$ of energy E_0 , the single-particle spectral function has an electron-addition and an electron-removal branch,

$$A_p(\omega) = \sum_n |\langle n^{N+1} | \hat{c}_p^\dagger | 0 \rangle|^2 \delta(\omega - (E_n^{N+1} - E_0)) + \sum_m |\langle m^{N-1} | \hat{c}_p | 0 \rangle|^2 \delta(\omega - (E_0 - E_m^{N-1})), \quad (1)$$

energies measured from E_0 ; the chemical potential $\mu = \frac{1}{2}(E_0^{N+1} - E_0^{N-1})$, which for the particle–hole-symmetric Hubbard model at half filling places the Fermi level mid-gap. We build the diagonalization subspace in three steps. (1) Prepare the response seeds $|\phi^\pm\rangle = \hat{c}_p^\dagger |0\rangle, \hat{c}_p |0\rangle$ in the $(N\pm 1)$ sectors. (2) Time-evolve each, $|\phi^\pm(t_k)\rangle = e^{-i\hat{H}t_k} |\phi^\pm\rangle$, and *sample computational-basis bitstrings*; the union of sampled configurations across the evolution times is the determinant subspace $\mathcal{S}^\pm$. (3) Form $\hat{H}_{\mathcal{S}^\pm} = P_{\mathcal{S}^\pm} \hat{H} P_{\mathcal{S}^\pm}$, diagonalize each as $\hat{H}_{\mathcal{S}^\pm} = \sum_m \tilde{E}_m^\pm |\tilde{m}^\pm\rangle \langle \tilde{m}^\pm|$, and evaluate the Lehmann sum inside $\mathcal{S}^\pm$. The retarded Green’s function is then assembled entirely on the classical processor,

$$G_{pq}(\omega) = \sum_m \frac{\langle 0 | \hat{c}_p | \tilde{m}^+ \rangle \langle \tilde{m}^+ | \hat{c}_q^\dagger | 0 \rangle}{\omega - (\tilde{E}_m^+ - E_0) + i\eta} + \sum_m \frac{\langle 0 | \hat{c}_q^\dagger | \tilde{m}^- \rangle \langle \tilde{m}^- | \hat{c}_p | 0 \rangle}{\omega - (E_0 - \tilde{E}_m^-) + i\eta}, \quad (2)$$

whose imaginary part is the Lorentzian-broadened $A_{pq}(\omega)$ at resolution η . Because the sampled configurations—the union over all orbital seeds $\{\hat{c}_q^\dagger | 0\rangle\}_q$ in the charged $(N\pm 1)$ sectors, and number-conserving density seeds $\{\hat{n}_q | 0\rangle\}_q$ in the neutral N sector—span the sector-specific subspaces in

which the *full* response matrices live, they yield without further measurement the momentum-resolved spectral function $A(k, \omega)$ (by Fourier transform of the charged G_{pq}) and the dynamical structure factor $S(q, \omega)$ (from the neutral density response). The quantum processor measures *only* bitstrings; no Hadamard test, controlled unitary, or off-diagonal overlap is estimated on the device (the sample–recover–reconstruct primitive is schematized in Fig. 14).

2.2 Relation to quantum subspace and Krylov methods

The construction is a sampling realization of the quantum subspace idea. In exact arithmetic the span of the time-evolved seeds $\{e^{-i\hat{H}t_k}|\phi\rangle\}$ coincides with the order- K polynomial Krylov space [13, 40], and diagonalizing $\hat{H}$ within a subspace containing this space is the route taken by quantum subspace expansion [41], quantum filter diagonalization [43], and real-time quantum Krylov algorithms [42], whose stability and error theory are now well characterized [44]. What distinguishes our method is that these states are used not to *estimate matrix elements*—the step that requires Hadamard tests or quantum equation-of-motion measurements in Green’s-function algorithms on hardware [45, 47–50]—but as a *sampling distribution* that selects a determinant subspace, exactly as in sample-based diagonalization [7, 8, 12], now lifted to the $(N\pm 1)$ manifold and to a frequency-resolved observable. Closest on lattice models, Nogaki *et al.* [19] apply symmetry-adapted SQD to lattice-model ground-state *energies*; we instead lift the sampled subspace to the $(N\pm 1)$ and neutral sectors to reconstruct the frequency-resolved single-particle and collective *response*. The device *readout* is therefore that of SQD—computational-basis sampling—while the circuits are the real-time-evolution circuits of sample-based Krylov diagonalization [12, 14], and the dynamical information is reconstructed classically through Eq. (2).

2.3 Convergence and sampling cost

Two questions govern reliability: does the subspace reproduce the exact spectrum, and how many shots are required to populate it? Both admit an answer. Writing each branch of A_p through its spectral moments—for the addition branch, $\mu_r^+ = \int \omega^r A_p^+(\omega) d\omega = \langle 0|\hat{c}_p(\hat{H} - E_0)^r \hat{c}_p^\dagger|0\rangle$, and analogously for removal ($\hat{c}_p^\dagger, \hat{c}_p$)—a subspace containing the order- K Krylov space of that branch reproduces its exact moments through order $2K+1$ —the Gauss-quadrature property of moment methods [53]—so the underlying unbroadened Stieltjes measure is moment-exact through order $2K+1$ in each branch (the broadened A_p inherits this as geometric L_1 convergence at fixed $\eta > 0$, though its own moments of order ≥ 2 diverge), and its residual L_1 error is bounded by the spectral weight outside the captured subspace. Under the empirically observed concentration of the dominant Lehmann residues on a number of configurations that grows polynomially in the target accuracy, a polynomial shot budget suffices to realize this subspace with high probability; we make this precise in Theorem 1 (App. A), in the spirit of the convergence guarantees established for sample-based Krylov diagonalization [12–14, 55]. In practice the reconstruction is validated *a posteriori* by the exact branch sum rules $\int A_p^+ d\omega = 1 - \langle \hat{n}_p \rangle$ and $\int A_p^- d\omega = \langle \hat{n}_p \rangle$ (unit total weight) and by self-consistency of the sampled subspace, so that at scales where an exact reference is unavailable the result remains internally consistent and weight-normalized. These zeroth-moment checks are not themselves an error bound; the genuine certificate is the error bound of Theorem 1(iii), $\|A - A_S\|_1 \leq 2\|\phi\|^2(1 - w_S) + C\rho^{-K_S}$, which controls the reconstruction once the captured weight $w_S \rightarrow 1$ and the captured Krylov order K_S is adequate—an essential property given that the classical-hardness of a specific instance is empirical rather than a theorem (Sec. 6).

3. One-body magic and its decoupling from the sampling cost $|\mathcal{S}|$

The value of a sampling method is set by whether the distribution it samples is classically hard, and for fermions that hardness has a precise algebraic origin. Circuits of free-fermion (matchgate)

gates are classically simulable in polynomial time [64], equivalently the classical simulability of non-interacting-fermion dynamics [65], with a Grassmann/covariance-matrix formulation via fermionic linear optics [66, 67]. The resource that leaves this manifold is fermionic *non-Gaussianity*: every pure non-Gaussian fermionic state, injected into a matchgate circuit, promotes it to universal quantum computation [68]—the fermionic counterpart of the non-stabilizerness (“magic”) of universal qubit computation [69, 70]. A ladder of measures makes this quantitative: Gaussian rank and extent [75], the non-Gaussian gate count that bounds phase-sensitive free-fermion simulation [73, 74], and covariance-matrix monotones computable from two-point functions [76, 78].

We work with the fermionic AntiFlatness (FAF) [71],

$$\mathcal{F}_k = 2n_o - \frac{1}{2} \text{tr}[(\Gamma^\top \Gamma)^k] = 2n_o - \sum_{j=1}^{2n_o} \lambda_j^{2k}, \quad \Gamma_{mn} = -\frac{i}{2} \langle [\gamma_m, \gamma_n] \rangle, \quad (3)$$

where Γ is the $4n_o \times 4n_o$ Majorana covariance matrix and $\{\lambda_j\} \in [0, 1]$ the singular values of its orthogonal (Youla) normal form—the moduli of its eigenvalue pairs $\pm i\lambda_j$; here $2n_o$ is the number of spin-orbital *modes* (twice the spatial-orbital count n_o), distinct from the electron number of the $(N \pm 1)$ sectors, and $\mathcal{F}_k$ ranges in $[0, 2n_o]$. $\mathcal{F}_k = 0$ if and only if the state is Gaussian, and $\mathcal{F}_k$ is invariant under Gaussian (free-fermion) unitaries by construction [71]. The index k is a hierarchy of correlator orders, efficiently computable and experimentally accessible from two-point functions [71, 76, 79].

One-body collapse. On the number-conserving states of electronic structure the anomalous correlators $\langle c_i c_j \rangle$ vanish, so Γ is fixed by the one-particle reduced density matrix γ and its singular values are $\lambda_i = |2g_i - 1|$, with $g_i \in [0, 1]$ the natural spin-orbital occupations. The $k = 1$ member then reduces to a single 1-RDM invariant,

$$\boxed{\mathcal{F}_1 = \sum_i [1 - (2g_i - 1)^2] = 4 \sum_i g_i(1 - g_i) = 4 \text{tr}[\gamma(1 - \gamma)]}, \quad (4)$$

where the sum runs over spin orbitals and $g_i \in [0, 1]$ are the natural spin-orbital occupations. This chain is exact on any number-conserving state. For *spin-restricted* natural orbitals ($g_{a\uparrow} = g_{a\downarrow} = n_a/2$) it equals $2N_u$, with $N_u = \sum_a n_a(2 - n_a)$ (spatial occupations $n_a \in [0, 2]$) the *quadratic* effectively-unpaired-electron index of Takatsuka–Fueno–Yamaguchi and of Staroverov–Davidson [112, 113] (distinct from the *linear* $\sum_a \min(n_a, 2 - n_a)$ index of Head-Gordon [114]); a cleanly singly-occupied spin orbital contributes 0 to $\mathcal{F}_1$ but 2 to $2N_u$, so on the five genuinely spin-polarized (open-shell) members of Table 1 ($\text{O}_2, \text{NO}, \text{CN}, \text{OH}, \text{BeH}$) the two differ and we report the exact $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ directly. The invariant $\text{tr}[\gamma(1 - \gamma)]$ is the Gigena–Rossignoli one-body-entanglement linear entropy [115]. We verify Eq. (4)—and, on the closed-shell members, its equality to $2N_u$ —to machine precision across the dissociation of N_2 and the full suite of Table 1. Algebraically Eq. (4) is an immediate number-conserving specialization of the antiflatness of Ref. [71], and its ensemble form— $\mathcal{F}_1$ proportional to the one-body purity deficit—was independently noted for interacting-integrable eigenstates in Ref. [72]; our point is not the algebra but the exact, state-by-state *dictionary* it makes explicit. Three literatures—fermionic magic, effectively-unpaired electrons, and one-body entanglement—thus name one quantity, and, being a function of the occupation spectrum alone, $\mathcal{F}_1$ inherits the orbital-rotation invariance of the eigenvalues of γ . (Outside number conservation the collapse fails: a paired Gaussian state hides fractional occupations in anomalous correlators and keeps $\mathcal{F}_1 = 0$.)

Decoupling from the cost. That invariance is exactly what severs $\mathcal{F}_1$ from the sampling cost. The cost of a sample-based method is the determinant support $|\mathcal{S}|_\varepsilon$: the number of fixed-basis Slater determinants that carry $1 - \varepsilon^2$ of the weight (concretely, 99% of the weight for the molecular suite and $\varepsilon = 0.05$ for the lattice families). Throughout, $|\mathcal{S}|$ without a subscript denotes this thresholded count; where the *raw* support—the determinants of nonzero amplitude—is meant we say so explicitly,

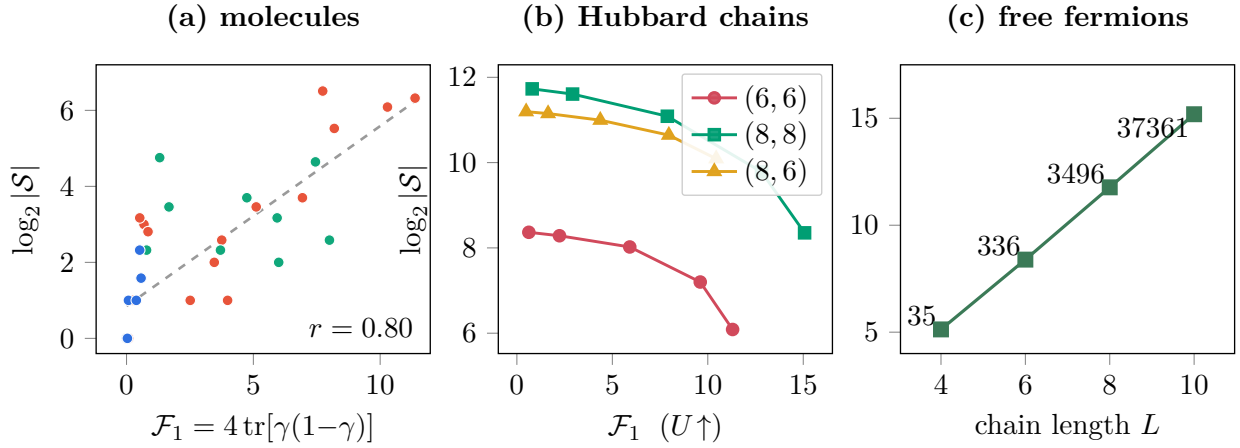


Figure 1. One-body magic $\mathcal{F}_1$ is decoupled from the sample-based determinant support $|\mathcal{S}|$. (a) Nineteen FCI-verified molecules at equilibrium and dissociation: in the static-correlation regime $\mathcal{F}_1$ and $|\mathcal{S}|$ rise together (Pearson $r = 0.80$ on $\log_2 |\mathcal{S}|$), yet $\mathcal{F}_1$ approaches its maximum while $|\mathcal{S}|$ stays a modest fraction of the active-space sector. (b) Hubbard chains at fixed filling (three representative sweeps; the full 30-point set is Fig. 2): increasing on-site repulsion U raises $\mathcal{F}_1$ but *lowers* the site-basis support $|\mathcal{S}|$ as the ground state localizes, inverting the sign of the correlation. (c) Free fermions ($U = 0$): a single Slater determinant with $\mathcal{F}_1 = 0$, whose site-basis support nonetheless grows as 35, 336, 3496, 37361 for chain length $L = 4, 6, 8, 10$. The $\mathcal{F}_1$ - $|\mathcal{S}|$ relationship is not sign stable—the signature of comparing an orbital invariant to a basis variant.

because the two do not obey the same bounds. Unlike $\mathcal{F}_1$, $|\mathcal{S}|$ is *basis dependent*—a single Slater determinant has $|\mathcal{S}| = 1$ in its own natural orbitals and exponentially many terms in a generic basis. An orbital rotation is a free (Gaussian) operation that leaves the occupation spectrum, hence $\mathcal{F}_1$, exactly fixed while changing $|\mathcal{S}|$ —from $O(1)$ in the natural-orbital basis to exponentially many determinants in a generic basis. A quantity invariant under operations that change $|\mathcal{S}|$ at fixed value cannot bound or predict it: one-body magic and the sampling cost are decoupled by symmetry.

The decoupling is two-sided, and its sign is not even stable (Fig. 1). Across the nineteen FCI-verified molecules of Table 1, $\mathcal{F}_1$ and $|\mathcal{S}|$ rise *together* in the static-correlation regime (Pearson $r = 0.80$ on $\log_2 |\mathcal{S}|$): bond dissociation drives fractional occupations and multireference weight in step. But a free-fermion ground state—a single Slater determinant, $\mathcal{F}_1 = 0$ —has $|\mathcal{S}|$ that grows exponentially in the site basis ($|\mathcal{S}| = 35, 336, 3496, 37361$ for chain length $L = 4, 6, 8, 10$), and along a Hubbard chain at fixed filling, increasing on-site repulsion *raises* $\mathcal{F}_1$ while *lowering* $|\mathcal{S}|$ as the ground state localizes onto few configurations. The relationship between an orbital invariant and a basis variant cannot be otherwise.

An obstruction, not a weakness. The failure is structural, not special to $\mathcal{F}_1$. For the *raw* determinant support (the unthresholded Slater rank), *no* continuous functional of a fixed set of reduced density matrices can lower-bound it (App. B): rank is a coefficient-insensitive count, so $|\Psi_\epsilon\rangle = |D_0\rangle + \epsilon \sum_{k=1}^K |D_k\rangle$ has all reduced density matrices converging to those of a single determinant—every continuous magic or correlation functional $\rightarrow 0$ —while the rank stays $\Theta(K)$. This continuity route does *not* by itself carry over to the operative, coefficient-*sensitive* $|\mathcal{S}|_\epsilon$, which collapses to 1 as $\epsilon \rightarrow 0$ in step with the functionals; the threshold-robust obstruction for $|\mathcal{S}|_\epsilon$ is supplied instead by the symmetry argument above—no orbital-rotation-invariant functional can bound it—and by two *equal-weight* witnesses, whose amplitudes do not shrink. The two-determinant cat $(|1 \cdots 10 \cdots 0\rangle + |0 \cdots 01 \cdots 1\rangle)/\sqrt{2}$ has $\gamma = \frac{1}{2}\mathbb{I}$, hence $\mathcal{F}_1 = 2n_o$ (maximal) yet Slater rank 2; and a paired (BCS) Gaussian state has Gaussian rank 1 yet fractional occupations. The only computable lower bounds on $|\mathcal{S}|$ are themselves basis dependent—the orbital-Schmidt rank across a bipartition, i.e. the tensor-network bond dimension [131], which we prove is a rigorous lower bound on the raw support, $\chi \leq |\mathcal{S}|$ (App. B, Prop. 2), and, consistently, the basis-dependent non-Gaussianity measures (such as the natural-orbital participation entropy) that *do* upper-bound the simulation cost [76]—

while a Gaussian-invariant magic measure is blind to them. We make the obstruction constructive (App. B.1, Thm. 2): an antisymmetrized product of K strongly orthogonal geminals carries one-body magic $\mathcal{F}_1 = 2N_u = 4K$, higher-order magic $\mathcal{F}_2 = 4K$, and pairing-basis determinant support $|\mathcal{S}| = 2^K$ all growing without bound, yet in that basis is a bond-dimension-2 tensor network— $O(1)$ bond dimension, $O(K)$ contraction and Born-sampling cost—so magic of *every* order is unbounded at fixed $O(1)$ bond dimension. Because $\mathcal{F}_k$ is orbital-rotation invariant while the sampling cost is not, no magic monotone is a sufficient certificate of hardness. Even the higher-order antiflatness $\mathcal{F}_{k \geq 2}$ fails: the witness has $\mathcal{F}_2 = 4K$ unbounded at $O(1)$ bond dimension. What is hard is *generic*, *high-rank* non-Gaussianity in the fixed operative basis—the connected higher-body cumulants, which no orbital-invariant $\mathcal{F}_k$ can capture—and not the paired stratum, whose amplitudes, overlaps, and number correlators are classically *estimable* to additive error under free-fermionic dynamics [77]. This bond dimension—which directly sets the classical cost of a matrix-product-state simulation [131]—is a rigorous lower bound on the *raw* support, $\chi \leq |\mathcal{S}|$ (Prop. 2). We are explicit that the quantity we measure is the operative thresholded support $|\mathcal{S}|_\varepsilon$, for which the relation is empirical rather than proved: the support and the Schmidt rank are truncated at different tolerances (99% of the weight for $|\mathcal{S}|$, $\varepsilon = 0.05$ for χ), so the thresholded support can fall below χ , and it does so in 7 of the 38 molecular points. What holds empirically is that, unlike the orbital-invariant $\mathcal{F}_1$, the bond dimension co-varies with the measured cost: across both the nineteen-molecule suite and open Hubbard chains, χ tracks $|\mathcal{S}|_\varepsilon$ (Spearman $\rho = 0.90$ over the $n = 38$ molecular equilibrium-and-dissociation points and $\rho = 0.57$ over the $n = 30$ Hubbard points; both $p < 0.01$). The molecular set by itself does not discriminate between the two candidate resources—one-body magic scores a comparable $\rho = 0.86$ on the identical $n = 38$ points, because there $\mathcal{F}_1$ and the support both merely track static correlation. What discriminates is the Hubbard interaction sweep, where χ holds at $\rho = 0.57$ while $\mathcal{F}_1$ collapses to $\rho = -0.11$ ($n = 30$, $p \approx 0.56$, not significant), increasing U raising $\mathcal{F}_1$ while lowering $|\mathcal{S}|_\varepsilon$; that contrast, not the molecular coefficient, is the evidence for the title. The absolute $|\mathcal{S}|_\varepsilon$, sector dimension D , and χ underlying these coefficients are tabulated per species in App. C. Many-body (bond-dimension) entanglement lower-bounds and, across the studied regimes, empirically tracks the sampling cost; one-body entanglement—equivalently the one-body magic $\mathcal{F}_1$ —does so only where it coincides with static correlation. That single-particle entanglement can be maximal while bipartite (bond) entanglement stays $O(1)$ is the fermionic counterpart of the magic–entanglement decoupling seen in qubit matrix-product states [132]. Figure 2 consolidates the picture over all 74 exact ground states of this work—nineteen molecules (equilibrium and dissociation), the Hubbard interaction sweep, and the chain/ladder pair. The determinant support tracks the bond dimension in *every* family (pooled Spearman $\rho = 0.72$, and positive within each), with the geminal witness ($\chi = 2$, $|\mathcal{S}| = 2^K$) the deliberately loose lower bound; one-body magic does not, and fails in two visibly distinct ways—the Hubbard branch runs the *wrong* way ($U \uparrow$ raises $\mathcal{F}_1$ while lowering $|\mathcal{S}|$) and the chain/ladder points sit at *identical* $\mathcal{F}_1$ with $|\mathcal{S}|$ differing by up to $\sim 2.2\times$. The cost is a property of the sampler’s basis, read by χ ; the orbital invariant $\mathcal{F}_1$ is blind to it.

What one-body magic does, and where the hardness is. None of this makes $\mathcal{F}_1$ uninformative. It is a faithful, efficiently measurable detector of *static* (multireference) correlation—generalizing to fermionic non-Gaussianity, and to nineteen FCI-verified molecules, the qubit non-stabilizerness peak of molecular bonding [81] and the reported proportionality between qubit non-stabilizerness and Hartree–Fock weight (correlation) for weak-to-moderate correlation—itself reported to break down beyond the Coulson–Fischer point [120] (Table 1). What it is *not* is a predictor of the classical *cost* of sampling, which—as the χ – $|\mathcal{S}|$ tracking above shows—is lower-bounded and tracked by the basis-dependent entanglement, and by the fixed basis a sampler cannot rotate, not by the orbital invariant $\mathcal{F}_1$. Cost and quantum hardness are thus distinct axes: the classical cost of the simulation is governed by entanglement, the bond dimension χ , whereas a genuine quantum advantage—a regime beyond *every* classical method—would require *generic*, *high-rank* non-Gaussianity in the operative basis, the connected two- and higher-body cumulants beyond the covariance [116], and specifically not the paired stratum, whose free-fermionic estimation primitives remain classically tractable [77]; the

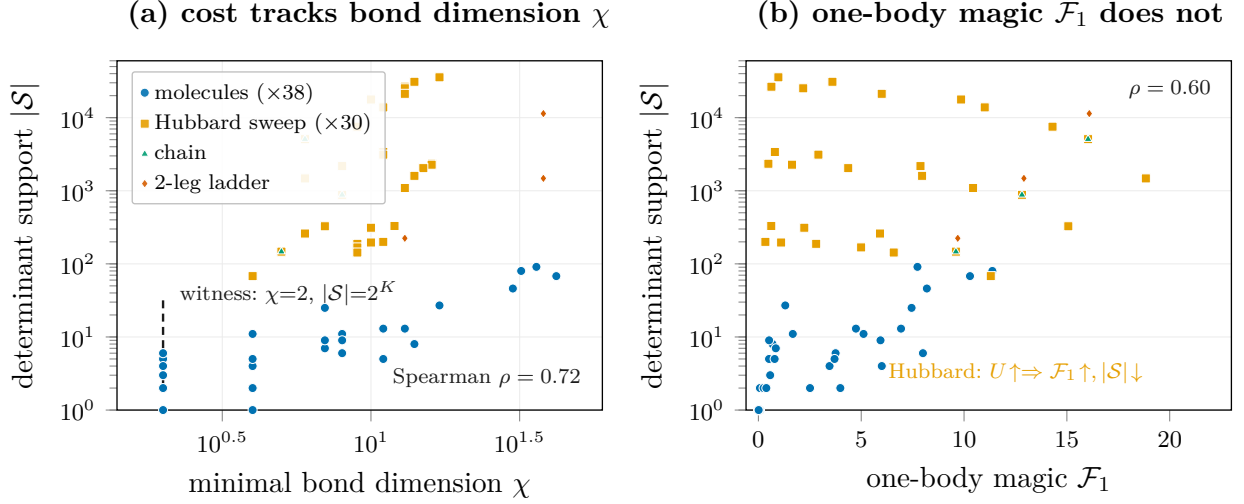


Figure 2. The resource thesis over 74 exact ground states. Nineteen molecules at equilibrium and dissociation, a Hubbard interaction sweep, and a matched chain/two-leg-ladder pair (all exact diagonalization). **(a)** The determinant support $|\mathcal{S}|$ tracks the minimal bond dimension χ in every family (pooled Spearman $\rho = 0.72$; the geminal witness $\chi=2$, $|\mathcal{S}|=2^K$ is the deliberately loose lower bound). **(b)** One-body magic $\mathcal{F}_1$ does not (pooled Spearman $\rho = 0.60$ for $\mathcal{F}_1$ vs $|\mathcal{S}|$ over the same 74 states): the Hubbard branch runs the wrong way ($U \uparrow \Rightarrow \mathcal{F}_1 \uparrow, |\mathcal{S}| \downarrow$) and the chain/ladder points share the same $\mathcal{F}_1$ at $|\mathcal{S}|$ differing by up to $\sim 2.2\times$. Cost is lower-bounded and tracked by the basis-dependent χ , not by the orbital-invariant $\mathcal{F}_1$.

orbital-invariant one-body magic $\mathcal{F}_1$ certifies neither. Fermionic magic thus supplies the missing link between the resource theory of non-Gaussianity and sample-based electronic-structure computation—but the link is a *demarcation*: what is so easily measured (invariant one-body magic) diagnoses correlation, while the cost lives in the entanglement and any genuine advantage one stratum above, in the generic higher-order sector.

4. Results

Having established analytically the resource that tracks the sampling cost (Sec. 3), we now demonstrate the method and that resource numerically. We validate the reconstruction against exact diagonalization, map the resource axis across lattices and a nineteen-molecule chemical suite, benchmark the time-evolution selector against standard classical selectors (conceding the strongest, heat-bath CI), and scale the demonstration toward the beyond-classical regime.

4.1 Validation

On Hubbard chains the bitstring-sampled subspace reproduces the exact multi-peak single-particle spectral function (Fig. 3). The relative- L_1 error falls toward zero as the time-evolution sampling discovers the spectral configurations, reaching 0.017 for the complete-subspace reconstruction (panel a; the 8-seed finite-shot curve of panel b flattens toward ≈ 0.019), and the addition-branch sum rule $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle$ is satisfied exactly—confirming that the classically-assembled Green’s function of Eq. (2) inherits the accuracy of the sampled subspace, not of any device-side estimation. The same primitive extends to the momentum-dependent $A(k, \omega)$ —whose dispersing lower and upper Hubbard bands and mid-gap Fermi level are shown *exactly* in Fig. 4—and to the dynamical structure factor $S(q, \omega)$ from a number-conserving density seed (Fig. 6): a single sampling primitive covers the photoemission and density-response channels that classically require separate dynamical-DMRG computations [89, 91]. From a spin seed the same primitive furnishes the spin structure factor $S^{zz}(q, \omega)$ (Fig. 7), and the two collective responses lay bare spin–charge separation in the two-particle channel: the charge continuum is *gapped* by the Mott gap Δ , whereas the spin continuum is *gapless*

(in the thermodynamic limit), its bandwidth set by the far smaller exchange $J = 4t^2/U$. At subspace saturation the bitstring-sampled reconstruction reproduces the exact result at every momentum to relative- L_1 below $\sim 0.5\%$ (Fig. 5: the sampled $A(k, \omega)$ overlays the exact target across the Brillouin zone, mean per-momentum error 2.2×10^{-3} , at a subspace of 85% of the $(N \pm 1)$ sector); crucially, however—and unlike the ground-state energy—saturating a *full* dynamical spectral function requires a substantial—though *shrinking*—fraction of the $(N+1)$ sector (0.92 at $L = 4$ down to 0.08 at $L = 14$, Fig. 12), because the spectral weight is distributed across the excited-state spectrum rather than concentrated in a few dominant configurations. This subspace-fraction cost, not any device-side accuracy, is the resource that Sec. 3 analyses. The continued-fraction reconstruction itself carries the standard Lanczos caveats—finite sampling of the subspace and loss of orthogonality at high recursion order can seed spurious low-weight poles—mitigated here by seed averaging and the finite broadening η .

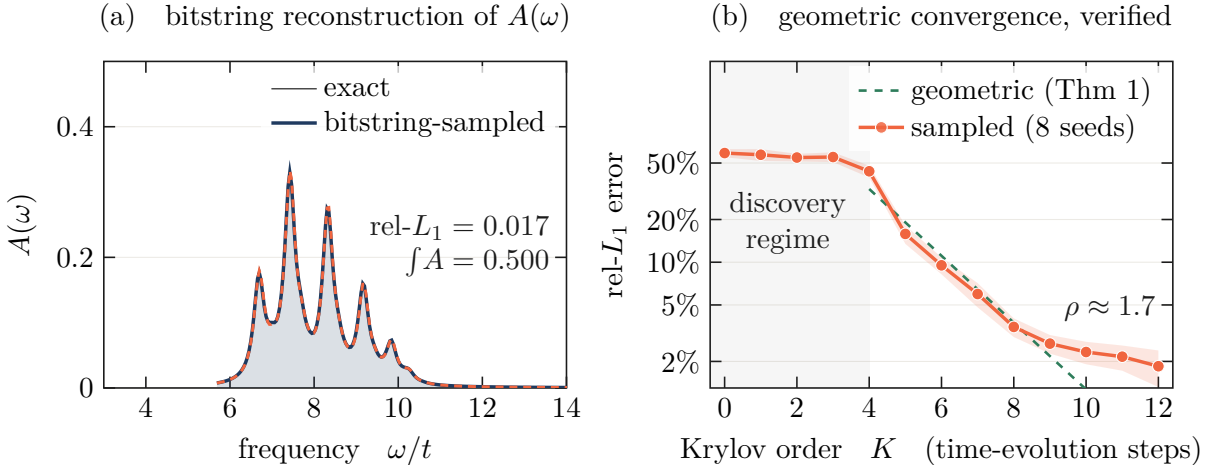


Figure 3. The method, validated—and its convergence matches the theorem. (a) The bitstring-sampled single-particle spectral function reproduces the exact $A(\omega)$ (Hubbard $L = 6$, $U/t = 8$; broadening $\eta = 0.15t$) to relative- L_1 0.017—the exact (filled) and reconstructed (dashed) curves are indistinguishable—and the reconstructed spectral weights obey the exact addition-branch sum rule $\int A^+ d\omega = 0.500 = 1 - \langle n_p \rangle$ to machine precision (the panel shows the addition branch $\omega > 0$; the full A integrates to 1). (b) Convergence of the relative- L_1 error with the Krylov order K (number of time-evolution steps sampled), seed-averaged over eight independent runs (band = standard deviation). Two regimes appear: a short *discovery* phase in which the time-evolution sampling first locates the spectral configurations, followed by a *geometric collapse* $\text{rel-}L_1 \sim \rho^{-K}$ with $\rho \approx 1.7$ —precisely the geometric convergence guaranteed by the moment-exactness theorem (Thm. 1, App. A). The sampled error tracks the geometric law until it flattens toward the 8-seed finite-shot value (≈ 0.019 at $K = 12$); the complete-subspace reconstruction of panel (a) reaches 0.017.

4.2 Fermionic magic marks the multireference regime

Having validated the dynamical primitive, we now turn from the observable to its resource. The same bitstring-sampled subspaces that reconstruct these spectra also let us evaluate, across the nineteen-molecule suite, the one-body fermionic magic $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ ($= 2N_u$ for the closed-shell members) identified in Sec. 3—and it tracks precisely the *multireference* character of the chemistry, not the sampling cost.

Across **nineteen molecules** spanning every bonding regime—covalent single, double, and triple (H_2 , F_2 , C_2 , N_2 , CO), polar and ionic (HF , LiF , LiH), multi-centre (H_2O , NH_3 , BeH_2), metal hydrides (BH , BeH), open-shell radicals (OH , CN , NO), the O_2 triplet, and hydrogen chains (H_4 , H_6)—we compute the fermionic AntiFlatness in a valence bond-breaking active space, **verifying every ground-state energy against exact active-space FCI to below 10^{-5} Ha** (Figs. 8, 9, Table 1). On the same suite the sampling method reproduces the exact (orbital-projected) spectral function $A(\omega)$ from bitstring-sampled subspaces to relative- L_1 error below 10^{-3} at every molecule—at the dissociation

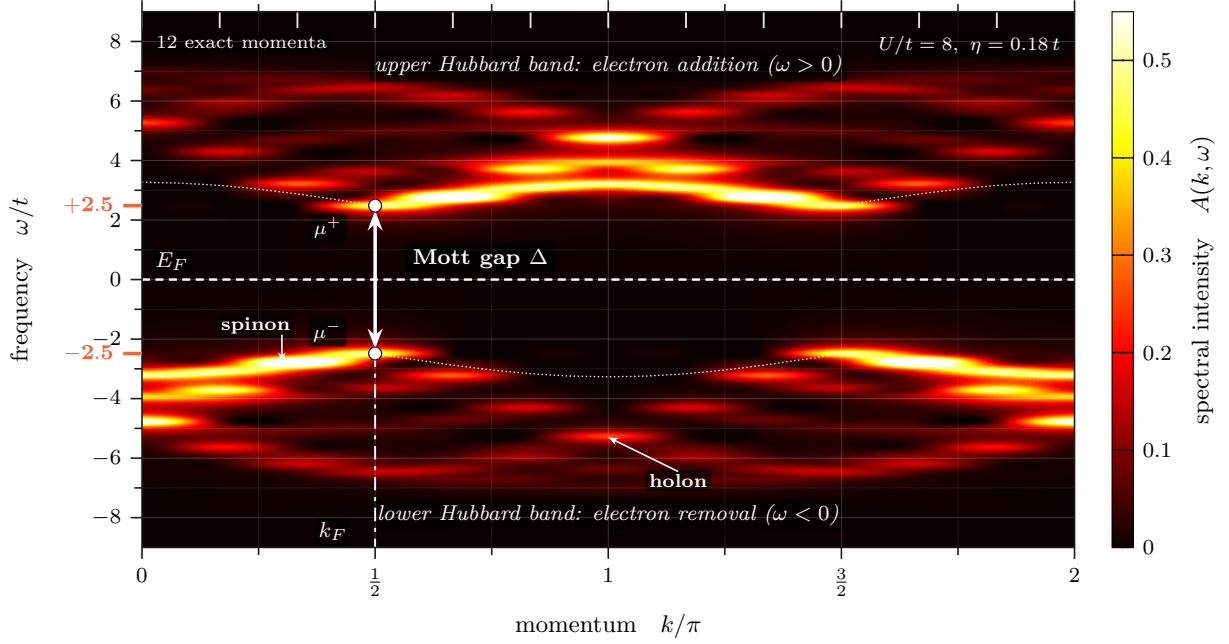


Figure 4. Momentum-resolved spectral function: the exact dynamical target. The single-particle spectral function $A(k, \omega)$ of the half-filled Hubbard chain ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18 t$), computed exactly from the resolvent $\langle \phi | (z - \hat{H})^{-1} | \phi \rangle$ via the Haydock continued fraction [54] over the full $(N \pm 1)$ sectors (equivalent to the Lehmann sum in exact arithmetic)—the frequency- and momentum-resolved observable that sample-based diagonalization targets but its energy-only output cannot itself deliver. The two dispersing Hubbard bands—electron removal (lower, $\omega < 0$; photoemission) and electron addition (upper, $\omega > 0$; inverse photoemission)—are separated by the correlation-driven *Mott gap* $\Delta = \mu^+ - \mu^- = 4.97 t$ between the highest occupied state (μ^- , lower edge) and the lowest unoccupied state (μ^+ , upper edge), obtained from the exact $(N \pm 1)$ ground-state energies (ticks at $\pm 2.484 t$); the broadened spectral peaks of the plotted $A(k_F, \omega)$ sit slightly wider, at $\pm 2.49 t$ (a $4.99 t$ splitting), the $\sim 0.02 t$ excess being the $\eta = 0.18 t$ Lorentzian broadening; the thermodynamic Lieb–Wu value [133] is $4.68 t$, the $L = 12$ excess being finite-size. The gap is *direct* at the Fermi momentum $k_F = \pi/2$. Particle–hole symmetry places the Fermi level at mid-gap ($\omega = 0$), verified to machine precision, and each momentum obeys the analytic spectral sum rule $\int A(k, \omega) d\omega = 1$ (the plotted broadened grid recovers it to $\sim 2\%$ within the $\pm 9t$ window). The lower Hubbard band is not a single quasiparticle band but a *spin–charge-separated* continuum: a narrow spinon branch (bandwidth $\pi J/2 = 0.79 t$ [134], $J = 4t^2/U$; dashed, validated against the exact peak dispersion to $< 0.1 t$) rides a broad holon (charge, $\sim 4t$) continuum—the hallmark fractionalization of the 1D Mott insulator [133]. The momentum axis is a periodic cubic-spline interpolation of the 12 physical momenta (ticks, top). Faithfully *sampling* this full momentum-resolved observable—as opposed to the ground-state energy—requires a large fraction of the $(N+1)$ sector (Fig. 12), the cost quantified in Sec. 3.

geometry, using a sampled subspace a fraction ≥ 0.10 of the $(N+1)$ sector, full only for the smallest ionic active spaces where the sector itself is tiny (Table 1)—so the reconstruction is both exact and efficient across chemistry. The magic picture is sharp, and it is *not* a universal bond-length effect. Magic *switches on* for covalent bond-breaking—rising from near-Gaussian at equilibrium to a large fraction of its maximum on stretch (F_2 , N_2 , H_2O , HF , NH_3 , CO). It is *already large at equilibrium* for the inherently multireference C_2 and O_2 , whose near-degenerate ground states carry static correlation before any bond is stretched. And it *stays near-Gaussian* for ionic and weakly-correlated bonds (LiF , LiH , BH , BeH), whose dissociation remains essentially single-configurational—the ionic–covalent avoided crossing of LiF lies well beyond the stretch shown. Fermionic magic therefore tracks *multireference*, *static-correlation* character, not bond length *per se*. This refines the qubit non-stabilizerness peak identified for molecular bonding [81] into a fermionic, Gaussian-referenced, FCI-verified statement: the fermionic AntiFlatness [71, 76]—efficiently computable from the Majorana covariance matrix, and a genuine magic monotone for matchgate circuits [68]—is a faithful one-body diagnostic of static-correlation character, reading *low* for the near-Gaussian LiF and LiH

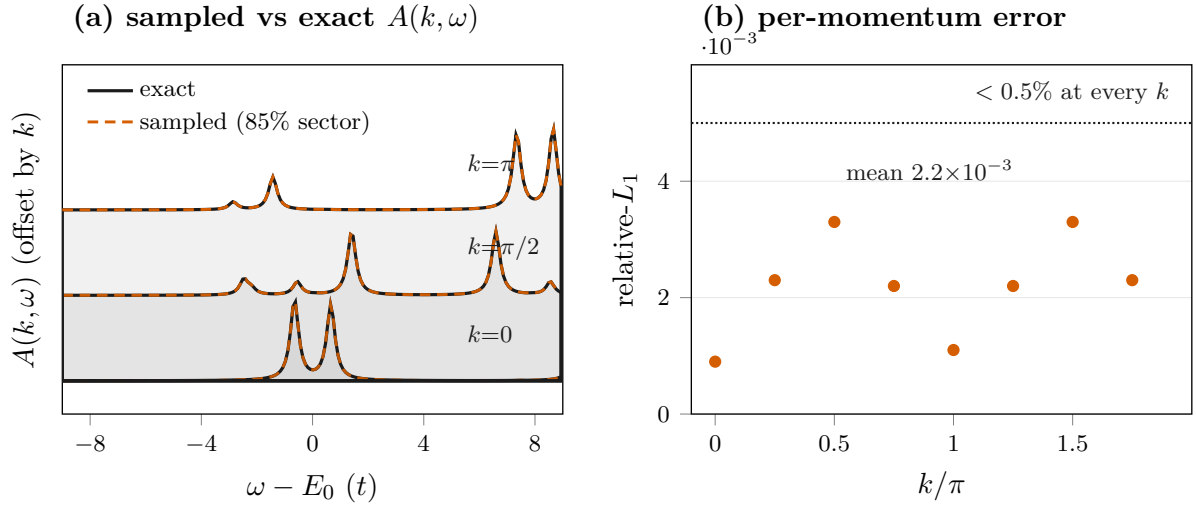


Figure 5. The momentum-resolved spectral function reconstructed from bitstring-sampled subspaces, overlaid on the exact target. (a) Bitstring-sampled $A(k, \omega)$ (dashed) versus the exact Haydock result (solid, shaded) at three representative momenta ($k = 0, \pi/2, \pi$; Hubbard chain, $L = 8$, $U/t = 8$, $\eta = 0.18 t$; curves offset vertically by k), reconstructed by time-evolution configuration sampling in the $(N \pm 1)$ sectors and the classical Lehmann assembly of Eq. (2)—not the exact target of Fig. 4, but the *sampled* output. (b) Relative- L_1 error of the sampled reconstruction at every physical momentum, at a subspace of 85% of the $(N \pm 1)$ sector: below 0.5% throughout (mean 2.2×10^{-3}). The reconstruction is faithful across the Brillouin zone, at the sector-fraction cost quantified in Sec. 3.

and *high* for stretched N_2 or F_2 . As Sec. 3 makes precise, this is a diagnostic of multireference character, not a predictor of the classical sampling cost: the two are decoupled, and the cost of these strongly-correlated targets is set by the higher-order structure the one-body magic does not resolve.

The N_2 triple-bond dissociation ties the picture together (Fig. 10): as the bond stretches, the sharp quasiparticle feature of the addition spectrum $A(\omega)$ broadens into a correlated multiplet and its integrated addition weight $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle$ falls from 0.98 to 0.52 as the frontier orbital fills toward half-occupation on homolytic dissociation. Along the same coordinate the fermionic magic rises as the *exact* identity $\mathcal{F}_1 = 2N_u$ [Eq. (4)]—one-body magic *is* the orbital-rotation-invariant multireference count, the effective number of unpaired electrons N_u from the natural-orbital occupations, which approaches its atomic limit of 6—the two triplet $N(^4S)$ atoms—as the bond fully breaks ($N_u = 5.9$ at the largest stretch shown; $\mathcal{F}_1^{\text{diss}} = 11.38$, i.e. $N_u = 5.69$, at the finite dissociation geometry of Table 1). The two curves in Fig. 10(b)—the frontier-orbital addition weight and the static-correlation magic—are distinct one-body diagnostics that cross as the bond breaks: the addition spectrum redistributes from a single quasiparticle into a multiplet precisely as multireference character switches on. Fermionic magic is thus a computable, basis-independent proxy for the strong static correlation that develops as the triple bond breaks.

4.3 The time-evolution selector versus classical Krylov and CIPSI selectors

The relevant question is not whether the sampled reconstruction is accurate—it is—but whether the time-evolution selector (the selection strategy a quantum device implements, evaluated here in classical simulation) discovers the important configurations more efficiently than deterministic classical selectors at fixed subspace size. Averaged over twelve independent sampling seeds, the time-evolution selector holds no robust advantage over classical polynomial Krylov and a configuration-interaction perturbation-theory (CIPSI [56, 57]) Green’s-function selector at small subspaces but moves ahead of both in the strong-coupling corner once the subspace exceeds ~ 120 determinants: at $U/t = 12$, $|\mathcal{S}| = 200$ it reaches relative- L_1 0.042 ± 0.004 against 0.729 for deterministic Krylov and 0.431 for CIPSI—a tenfold reduction (Fig. 11)—because deterministic Krylov plateaus, unable to resolve the delocalized configurations the strong- U response demands, whereas the time-evolution

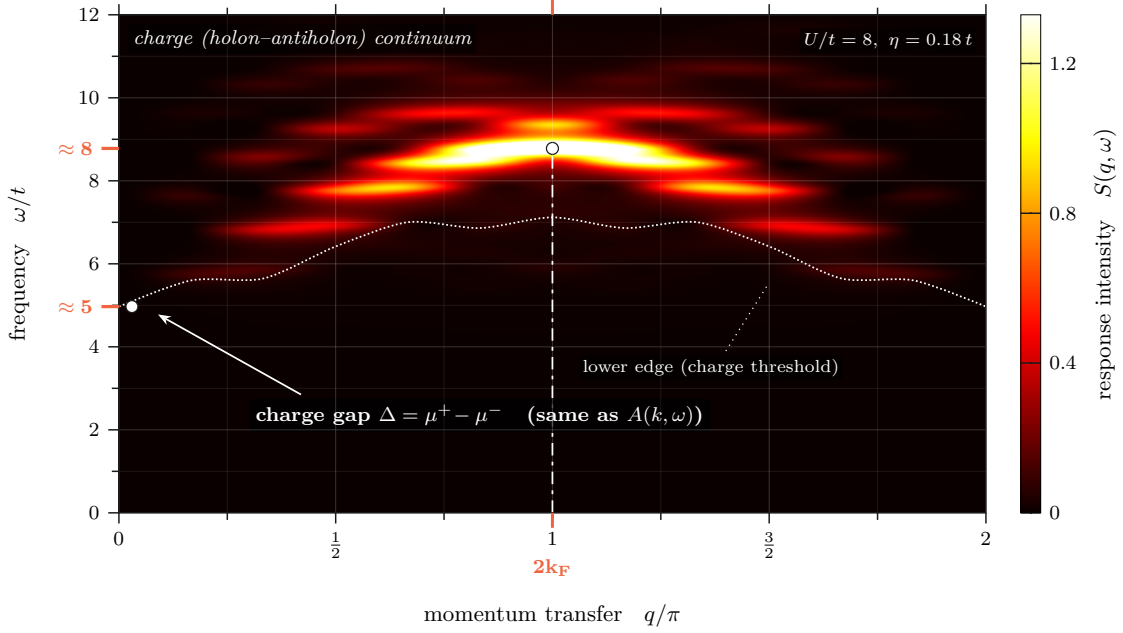


Figure 6. Dynamical structure factor: the exact density response. $S(q, \omega)$ of the half-filled Hubbard chain ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18t$)—the density–density response, a number-conserving (neutral) excitation, computed exactly via the Haydock continued fraction in the N -particle sector. Colour is the *response intensity* $S(q, \omega)$ (units $1/t$): bright means the system responds strongly at that (q, ω) , dark means it does not. The weight forms the two-particle charge (holon–antiholon) continuum, whose lower edge (dotted, the charge threshold) sets the gap—as $q \rightarrow 0$ it approaches Δ —and traces the lower-boundary dispersion of the continuum; its diffuse upper boundary is left unmarked. Two features are decisive. (i) The continuum is *gapped*: as $q \rightarrow 0$ its lower edge approaches the charge gap $\Delta = \mu^+ - \mu^- = 4.97t$ (exact $(N \pm 1)$ ground-state energies)—the *same* Mott gap that separates the Hubbard bands in the single-particle channel (Fig. 4): two independent dynamical observables, one correlation gap. (ii) The continuum peaks at $q = 2k_F = \pi$ at $\omega \approx U \approx 8t$, the bare Hubbard scale of a doubly-occupied site. The reflection $S(q) = S(2\pi - q)$ holds to $\sim 10^{-4}$ (Lanczos precision), and the analytic f-sum rule $\int \omega S(q, \omega) d\omega \propto (1 - \cos q)$ is recovered by the plotted (finite-window) grid to $\sim 0.3\%$. This is the collective observable the same bitstring-sampled primitive targets; as for $A(k, \omega)$, faithfully sampling the *full* response carries the sector-fraction cost of Sec. 3, where classically it needs a separate dynamical-DMRG computation.

selector reaches low error with fewer discovered configurations. The same picture holds against budget-limited real-time adaptive-sampling configuration interaction [100]. At small subspace size there is *no* robust advantage: the win is over these two deterministic selectors and is asymptotic in subspace size and coupling, not a small-scale artifact. We make no claim against the strongest classical selector: heat-bath configuration interaction remains competitive with sample-based selection at accessible sizes [25, 94, 95], consistent with our own companion review [27]. What drives the separation is the determinant count itself: the number of configurations over which the strong- U response spreads—and, with it, the extended-matchgate circuit extent [21]—grows with the evolution time the processor runs, while the quantum sampling stays polynomial in the target subspace size $|\mathcal{S}|$. The fermionic magic grows alongside but, as Sec. 3 establishes, does not itself set this cost.

4.4 Scaling of the sampling cost with system size

Across $L = 4$ –14 (8 to 28 qubits—the three largest, 20–28 qubits, reached by matrix-free Lanczos/Haydock beyond the dense-diagonalization frontier and validated against it at $L \leq 8$; these lattices are periodic rings, in contrast to the open chains of Sec. 3) two trends coexist (Fig. 12): the fermionic magic of the correlated ground state grows with system size ($\mathcal{F}_1 = 6.6 \rightarrow 9.4 \rightarrow 12.8 \rightarrow 16.0 \rightarrow 19.3 \rightarrow 22.5$)—but only *extensively*: per site it is flat ($\mathcal{F}_1/L = 1.65, 1.56, 1.60, 1.60, 1.61, 1.61$), so a fixed amount of static correlation accumulates with the sites and does not strengthen with

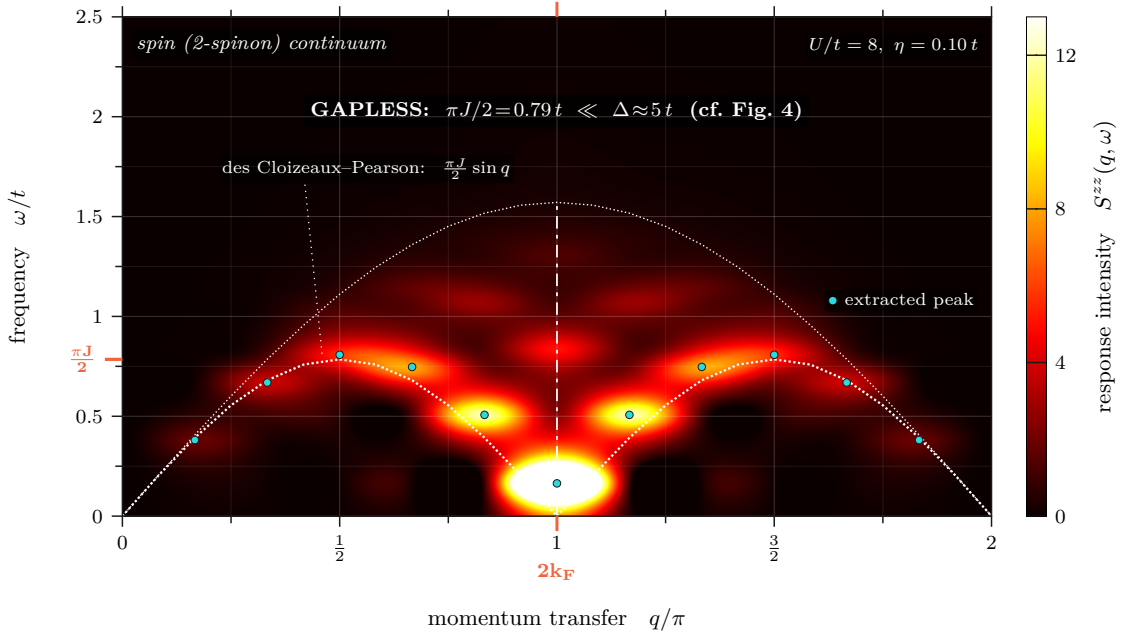


Figure 7. Spin structure factor: the gapless companion of the charge response. The dynamical spin structure factor $S^{zz}(q, \omega)$ of the same half-filled Hubbard chain ($U/t = 8$, $L = 12$; here $\eta = 0.10 t$, finer because the spin scale is small), computed exactly by the Haydock continued fraction from a spin seed $S_q^z|\psi_0\rangle$ in the N -particle sector. *Contrast Fig. 6:* the charge response is *gapped* by the Mott gap $\Delta \approx 5 t$, but the spin response is *gapless*—the two-spinon continuum touches $\omega = 0$ at $q = 0$ and $q = 2k_F = \pi$. This is spin-charge separation in the collective (two-particle) channel: charge and spin live in energetically separate worlds, the spin set by the superexchange $J = 4t^2/U = 0.5 t$, some $16\times$ smaller than U . The spectral weight piles up on the des Cloizeaux-Pearson [134] lower boundary $\frac{\pi J}{2}|\sin q|$ (dotted), bounded above by $\pi J|\sin \frac{q}{2}|$ [135]; the numerically-extracted peak dispersion (cyan markers) tracks this lower boundary to $\sim 3\%$; the antiferromagnetic soft mode at $q = \pi$ carries the dominant weight. The reflection $S^{zz}(q) = S^{zz}(2\pi - q)$ holds to machine precision and $S^{zz}(q \rightarrow 0) \rightarrow 0$ (total- S^z conservation). The same bitstring-sampled primitive that yields $A(k, \omega)$ and $S(q, \omega)$ targets this collective response as well.

size—while the $(N+1)$ -sector subspace *fraction* needed to reconstruct $A(\omega)$ to relative- $L_1 < 0.05$ *falls* steeply ($0.92 \rightarrow 0.82 \rightarrow 0.56 \rightarrow 0.36 \rightarrow 0.17 \rightarrow 0.08$, i.e. from 92% to 8%), the favourable face of the polynomial-shot bound of Theorem 1 (App. A) and consistent with the convergence behaviour established for sample-based Krylov diagonalization [12, 14]. The absolute subspace $|\mathcal{S}|$ nonetheless grows—indeed, for these lattice models it grows *exponentially* with system size, an intrinsic consequence of the computational-basis delocalization of the ground state [26]. Quantitatively, a least-squares fit over $L = 4\text{--}14$ gives $|\mathcal{S}| \sim e^{1.05 L}$ ($R^2 = 0.998$)—exponential, but *slower* than the full $(N+1)$ sector $D \sim e^{1.30 L}$, so the required fraction contracts as $\sim e^{-0.25 L}$ (halving every ~ 2.8 sites, $R^2 = 0.94$). Figure 12 is therefore not a claim of small *absolute* cost but the complementary, quantitative statement that the *fraction* of the sector required for a fixed spectral accuracy shrinks exponentially. And what tracks that absolute size is not the magic but the entanglement, as a direct chain-versus-ladder test makes concrete (Fig. 13). At *matched* one-body magic—a two-leg Hubbard ladder [19] and a chain of the same site count have near-identical $\mathcal{F}_1$ (12.82 vs 12.91 at 16 qubits, half filling, $U/t = 8$), since $\mathcal{F}_1$ is a geometry-blind one-body invariant—the ladder’s perpendicular cut carries an area-law boundary of two legs rather than one, so its minimal bond dimension is far larger ($\chi = 8 \rightarrow 38$ at 16 qubits; the chain-to-ladder ratio reads $\times 2.6, \times 4.8, \times 6.3$ at 12, 16 and 20 qubits, the ladder χ saturating at 38 while the chain χ is non-monotone (5, 8, 6), so the last step arises from the chain falling $8 \rightarrow 6$ rather than from ladder growth) and its response-targeted $(N\pm 1)$ fraction correspondingly higher ($0.43 \rightarrow 0.63$). The cost is lower-bounded and tracked by the bond dimension χ , not by $\mathcal{F}_1$ —the decoupling of Sec. 3, now exhibited as a controlled contrast in which the classical cost tracks χ and not $\mathcal{F}_1$. We are explicit about scope: a two-leg

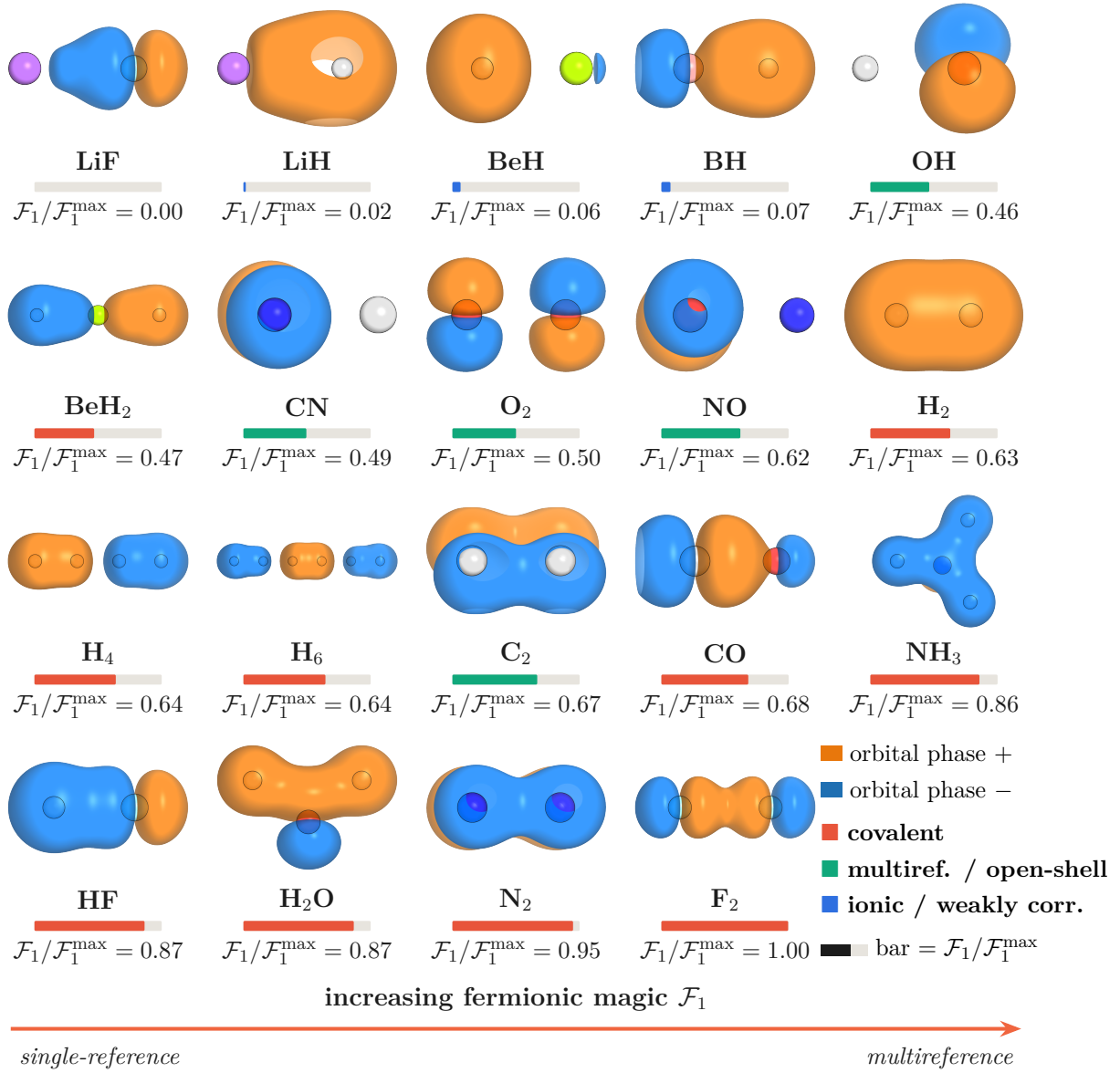


Figure 8. Nineteen molecules ordered by fermionic magic—the multireference axis. The frontier (HOMO) molecular orbital of each molecule at its dissociation geometry (PyMOL ray-traced two-phase isosurface; orange/blue = orbital phase $\pm$; CPK atoms), arranged left-to-right, top-to-bottom in order of increasing FCI-verified fermionic magic (single-reference $\rightarrow$ strongly multireference). The bar beneath each panel has length $\mathcal{F}_1/\mathcal{F}_1^{\max}$ and colour set by bonding regime; molecule names are in neutral dark ink. The ordering makes the diagnostic visual: ionic and weakly-correlated bonds (LiF, LiH, BeH, BH) sit at the near-Gaussian, single-reference end; covalent bond-breaking (HF, H₂O, N₂, F₂) at the strongly-multireference end; the inherently multireference C₂ and O₂ in between. This one-body magic faithfully diagnoses static-correlation character; it is decoupled from the sampling cost $|\mathcal{S}|$ (Sec. 3, Fig. 1). Orbitals are the mean-field frontier MO; the magic ordering is the correlated, FCI-verified quantity.

ladder with $\chi \approx 38$ is quasi-one-dimensional and remains near-exact for (dynamical) density-matrix renormalization group [83, 89]—it is *not* itself beyond the tensor-network frontier, which lies in genuinely two-dimensional (wide-cylinder) geometries and the all-to-all orbital graphs of quantum chemistry. What the ladder isolates is the *resource* statement—that entanglement, not one-body magic, lower-bounds and tracks the determinant support. Because χ is precisely the bond dimension a (dynamical) density-matrix renormalization-group calculation must carry [83, 89], panel (a) is a direct classical-cost comparison: the $\times 4.8$ chain-to-ladder gap at fixed $\mathcal{F}_1$ (16 qubits) is a gap in the tensor-network spectral cost itself, and by Prop. 2 it is a rigorous lower bound on the raw

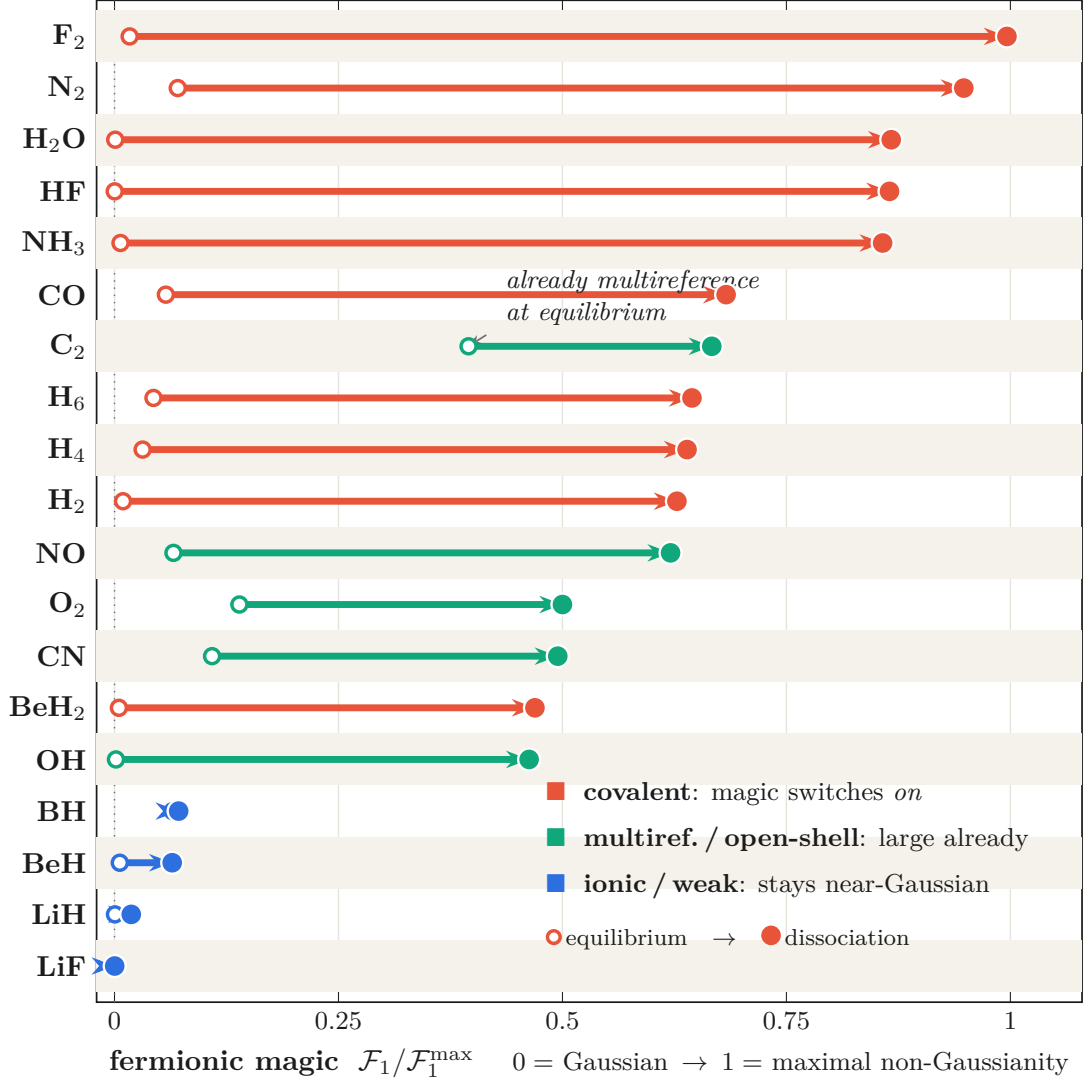


Figure 9. Fermionic magic marks the multireference regime across nineteen molecules. Normalized fermionic AntiFlatness $\mathcal{F}_1/\mathcal{F}_1^{\max}$ (0: Gaussian/mean-field; 1: maximal non-Gaussianity) at equilibrium (open circles) and at dissociation (filled), for all nineteen molecules, sorted by dissociation magic and coloured by regime. Covalent bond-breaking switches magic on; the inherently multireference C₂ and O₂ are already large at equilibrium; ionic and weakly-correlated bonds (LiF, LiH, BH, BeH) stay near-Gaussian. Every ground-state energy is verified against exact active-space FCI to $< 10^{-5}$ Ha (Table 1); computed in valence bond-breaking active spaces, cc-pVDZ.

determinant support—a resource contrast, with no beyond-classical claim asserted.

5. Hardware demonstration

The method runs on today’s processors without realizing the number-non-conserving operator $\hat{c}_p^\dagger$ on the device. One prepares a *number-conserving* $(N+1)$ -electron reference—a Slater or local-unitary-cluster-Jastrow state with $n_\alpha = N/2 + 1$ —and Trotter-evolves it shallowly, mapping on-site interactions to native fractional R_{ZZ} gates and hopping to Givens rotations (Fig. 14). Computational-basis bitstrings are sampled, recovered to the correct particle-number sector by self-consistent configuration recovery [7], and the entire $\hat{c}_p^\dagger$ /Lehmann construction of Eq. (2) is assembled classically to yield the full matrix $A_{pq}(\omega)$. Only a few Trotter steps are needed, because the classical diagonalization is variational in the sampled subspace.

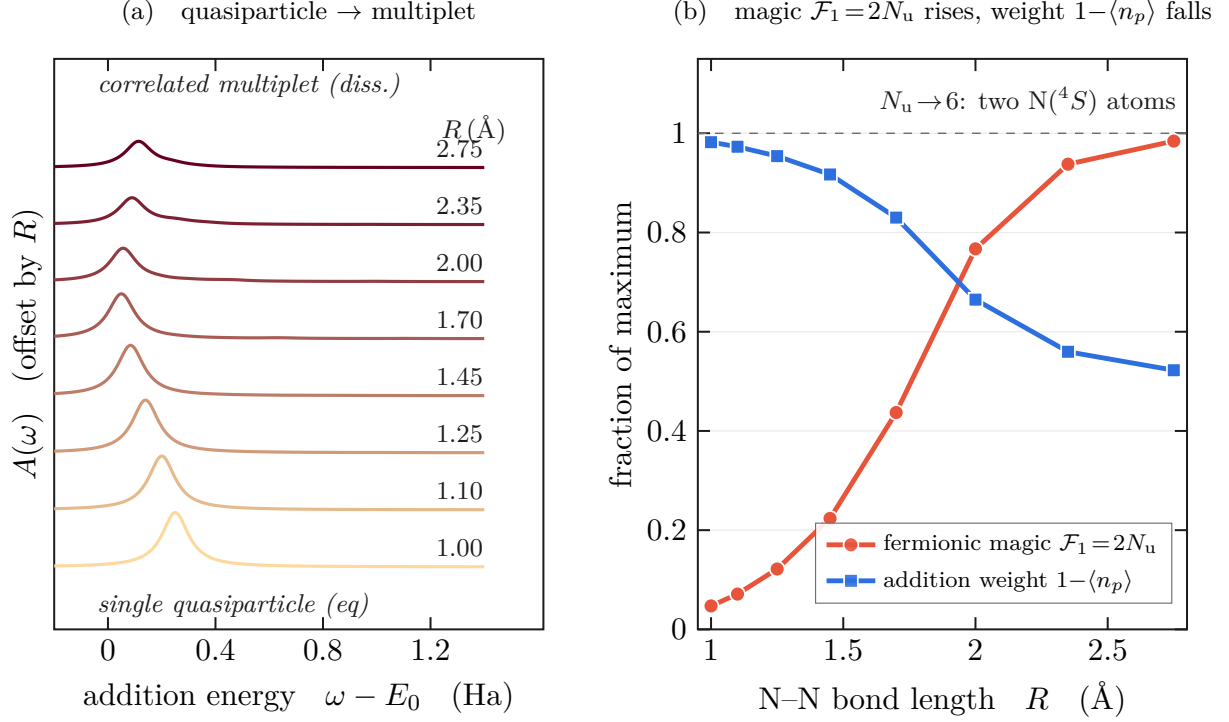


Figure 10. Magic and multireference character in N_2 dissociation. (a) The exact addition spectral function $A(\omega)$ (orbital-projected, CAS(6,6), cc-pVDZ) along the triple-bond stretch, offset by R : the sharp coherent quasiparticle peak at equilibrium ($R = 1.0$ Å) broadens into a correlated multiplet on dissociation as its spectral weight collapses. (b) Two physically independent quantities along the same coordinate. The fermionic magic obeys the *exact* identity $\mathcal{F}_1 = 2N_u$ [Eq. (4), verified to $< 10^{-13}$], so a single curve carries both: it rises with the orbital-rotation-invariant multireference count—the effective number of unpaired electrons $N_u = \sum_i n_i(2 - n_i)$ from the natural-orbital occupations n_i , approaching its atomic limit of 6 (the two triplet $N(^4S)$ atoms) as the bond fully breaks. The integrated addition weight $\int A^+ d\omega = 1 - \langle \hat{n}_p \rangle$ (from panel a) falls from 0.98 to 0.52 over the same stretch; the two cross as the bond breaks. Every geometry is FCI-verified to $< 10^{-5}$ Ha.

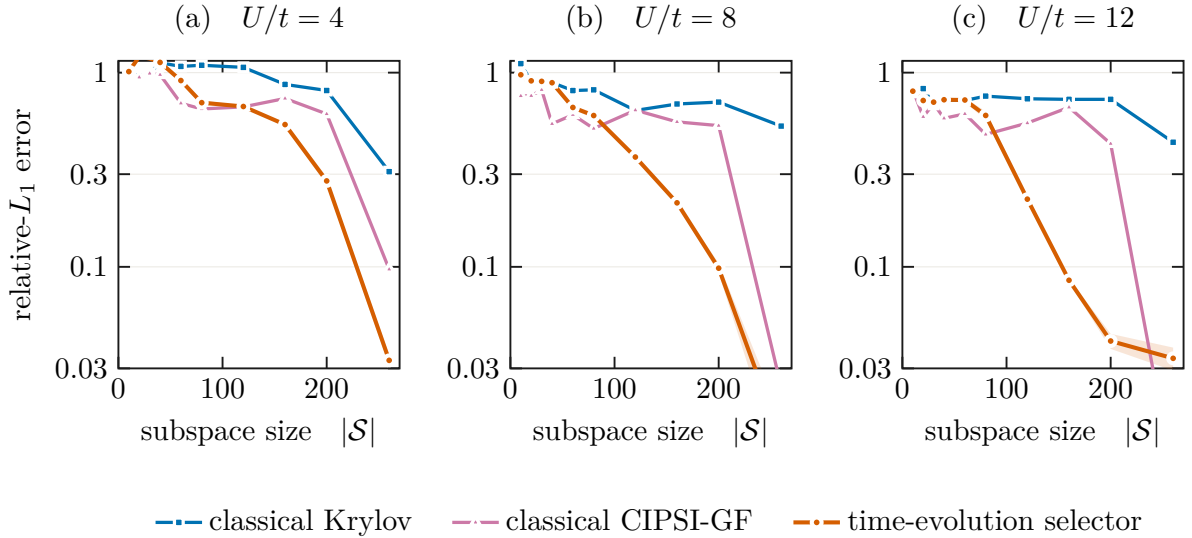


Figure 11. Time-evolution selector vs. classical Krylov and CIPSI selectors, seed-averaged (bands = seed spread; all curves evaluated in classical simulation). At strong coupling and $|\mathcal{S}| \gtrsim 120$ the time-evolution selector reaches low error with fewer discovered configurations, while classical Krylov plateaus and the CIPSI perturbative Green's-function selector trails; at small $|\mathcal{S}|$ there is no robust advantage.

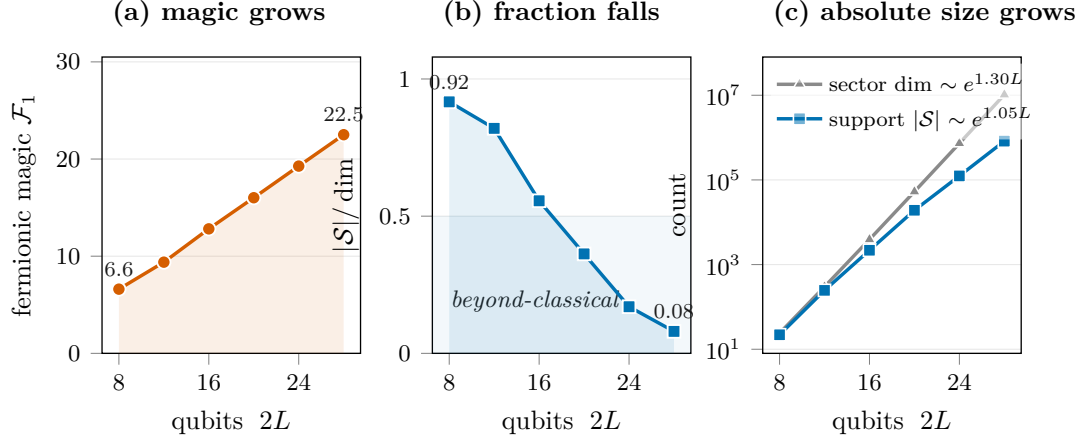


Figure 12. Scaling of the sampling cost with system size ($L = 4\text{--}14$; 8–28 qubits, half-filled *periodic* Hubbard rings at $U/t = 8$; the three largest sizes (20–28 qubits) computed by matrix-free Lanczos/Haydock, validated against dense diagonalization at $L \leq 8$). (a) The fermionic magic $\mathcal{F}_1$ of the correlated ground state grows with system size (6.6 $\rightarrow$ 22.5), extensively— $\mathcal{F}_1/L$ is flat at ≈ 1.6 , so this is accumulation, not strengthening, of static correlation. (b) The $(N+1)$ -sector subspace fraction $|S|/\dim$ needed to reconstruct $A(\omega)$ to relative- $L_1 < 0.05$ *falls* steeply (0.92 $\rightarrow$ 0.08)—the sampled reconstruction becomes relatively cheaper as the state grows harder to represent classically. (c) The absolute sizes nonetheless both grow *exponentially* (log scale): the determinant support $|S| \sim e^{1.05L}$ and the full $(N+1)$ sector dimension $\dim \sim e^{1.30L}$; the faster-growing sector is precisely why the *fraction* in (b) contracts. Fraction down, absolute cost up—the honest two-sided picture.

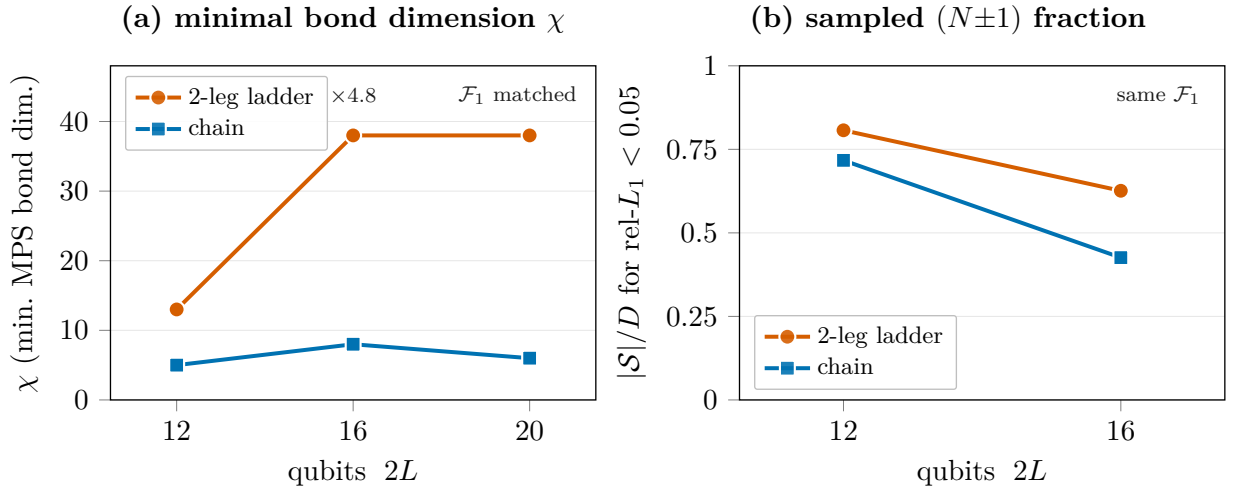


Figure 13. The cost is tracked by entanglement, not one-body magic—a controlled chain-versus-ladder contrast. A two-leg Hubbard ladder versus a chain of the *same* site count (half filling, $U/t = 8$; exact diagonalization). At matched fermionic magic $\mathcal{F}_1$ (open chain vs. ladder: 9.60/9.69, 12.82/12.91, 16.03/16.09 at 12/16/20 qubits— $\mathcal{F}_1$ is a geometry-blind one-body invariant) the ladder’s perpendicular cut has an area-law boundary of two rather than one. (a) Its minimal matrix-product-state bond dimension χ is far larger; the chain-to-ladder ratio grows over 12 $\rightarrow$ 20 qubits ($\times 2.6, \times 4.8, \times 6.3$), the ladder χ saturating at 38 while the chain χ stays small and does *not* grow (5, 8, 6 at 12/16/20 qubits—non-monotone, with no systematic increase over this range, consistent with the at most logarithmic entanglement growth of the gapless spin sector)—the reported χ are threshold-truncated Schmidt ranks. (b) The response-targeted $(N+1)$ fraction needed for spectral relative- $L_1 < 0.05$ is correspondingly higher for the ladder (these union $(N+1)$ response fractions differ from the single-sector $(N+1)$ $A(\omega)$ fractions of Fig. 12, a different observable). Same one-body magic, higher cost: the determinant support is lower-bounded and tracked by the basis-dependent bond dimension χ (Sec. 3), not by $\mathcal{F}_1$. A ladder of this width ($\chi \approx 38$) remains classically near-exact for dynamical DMRG; it is a resource contrast, not a beyond-classical claim.

Table 1. The nineteen-molecule fermionic-magic suite, computed and **verified against exact active-space FCI to $< 10^{-5}$ Ha at every geometry** (cc-pVDZ). The fermionic AntiFlatness $\mathcal{F}_1$ ($k=1$ Majorana antiflatness; $\mathcal{F}_1=0$ iff Gaussian/mean-field, maximum $\mathcal{F}_1^{\max}=2n_o$) is listed near equilibrium and at dissociation, in a valence bond-breaking active space $\text{CAS}(n_e, n_o)$. Fermionic magic switches on for covalent bond-breaking, is already large at equilibrium for inherently multireference species (C_2 , O_2), and stays near-Gaussian for ionic and weakly-correlated bonds—so $\mathcal{F}_1$ marks the multireference, strongly-correlated regime rather than bond length *per se*. The last two columns validate the *method*: at the dissociation geometry the bitstring-sampled reconstruction of the (orbital-projected) spectral function $A(\omega)$ matches the exact Lehmann result to relative- L_1 error, using a sampled subspace that is a fraction $|\mathcal{S}|/D$ of the full $(N+1)$ sector. The listed $\mathcal{F}_1 = 4 \text{tr}[\gamma(1-\gamma)]$ are directly computed spin-orbital values (for the five open-shell species they are *not* $2N_u$; see Sec. 3). Per-molecule equilibrium and dissociation geometries, active-space orbital indices, FCI energies and residuals, and the absolute $|\mathcal{S}|$, sector dimension D and minimal bond dimension χ are listed in App. C. †: near-degenerate ground state.

Molecule	Bonding	$\text{CAS}(n_e, n_o)$	$\mathcal{F}_1^{\text{eq}}$	$\mathcal{F}_1^{\text{diss}}$	$\mathcal{F}_1^{\text{diss}}/\mathcal{F}_1^{\text{max}}$	rel- L_1	$ \mathcal{S} /D$
<i>Covalent bond-breaking</i>							
F_2	covalent single	(1+1,2)	0.068	3.985	1.00	$< 10^{-13}$	0.50
N_2	covalent triple	(3+3,6)	0.847	11.377	0.95	2×10^{-5}	0.21
H_2O	multicentre	(2+2,4)	0.007	6.936	0.87	$< 10^{-13}$	0.50
HF	polar single	(1+1,2)	0.001	3.460	0.87	$< 10^{-13}$	1.00
NH_3	multicentre	(3+3,6)	0.080	10.289	0.86	1×10^{-5}	0.47
CO	polar triple	(3+3,6)	0.688	8.192	0.68	1×10^{-4}	0.41
H_6	H-chain	(3+3,6)	0.521	7.735	0.64	2×10^{-5}	0.50
H_4	H-chain	(2+2,4)	0.252	5.113	0.64	$< 10^{-13}$	0.50
H_2	covalent single	(1+1,2)	0.038	2.511	0.63	$< 10^{-13}$	0.50
BeH_2	multicentre	(2+2,4)	0.039	3.754	0.47	$< 10^{-13}$	0.25
<i>Multireference / open-shell</i>							
$\text{C}_2^\dagger$	covalent double	(4+4,6)	4.743	8.001	0.67	$< 10^{-13}$	0.10
NO	polar (radical)	(4+3,6)	0.790	7.449	0.62	2×10^{-6}	0.25
O_2	covalent double (triplet)	(5+3,6)	1.673	6.000	0.50	1×10^{-5}	0.25
CN	polar (radical)	(4+3,6)	1.307	5.938	0.49	1×10^{-5}	0.24
$\text{OH}^\dagger$	polar (radical)	(3+2,4)	0.013	3.703	0.46	$< 10^{-13}$	0.33
<i>Ionic / weakly correlated</i>							
BH	metal hydride	(2+2,4)	0.518	0.573	0.07	$< 10^{-13}$	0.25
BeH	metal hydride (radical)	(2+1,3)	0.035	0.386	0.06	$< 10^{-13}$	0.67
LiH	metal hydride	(1+1,2)	0.002	0.075	0.02	$< 10^{-13}$	1.00
LiF	ionic	(1+1,2)	0.000	0.001	0.00	$< 10^{-13}$	1.00

Executed on the IBM Heron processor `ibm_fez` for the $L = 6$ Hubbard model at $U/t = 4$ (broadening $\eta = 0.15t$; 12 qubits; the $t = 0$ reference plus $K = 7$ real-time-evolution circuits, whose 50,000 pooled computational-basis shots define the union subspace; TREX readout twirling, Pauli twirling, and dynamical decoupling, and *no* zero-noise extrapolation), the bitstring-sampled spectral function reproduces the exact $A(\omega)$ to within the line-broadening resolution (Fig. 15)—a demonstration on a real quantum computer that the open-gap method works. At this scale the $(N+1)$ sector is small enough that hardware sampling with self-consistent recovery captures essentially the complete subspace; the robustness underlying this clean result is quantified under a per-qubit bit-flip channel: self-consistent configuration recovery holds the spectral reconstruction near relative- $L_1 \approx 0.05$ out to $\varepsilon \approx 16\%$, far beyond the naive-post-selection range where the error grows steadily with the flip rate (Fig. 16). The demonstration realizes time-evolved QSCI [10] and sample-based Krylov diagonalization [12–14] lifted into the $(N\pm 1)$ sectors; the nearest published sibling also builds a subspace from sampled configurations—from short-time quantum-evolved states with classically projected long-time dynamics—but targets the *neutral*-sector molecular response, not the charged- $(N\pm 1)$ single-particle Green’s function our Lehmann construction assembles [51]; the hardware Green’s-function and dynamical-structure-factor results of the trapped-ion line [39, 49] use non-subspace routes. A ready-to-run notebook for the N_2 molecular spectral function—a double-factorized Trotter of the molecular Hamiltonian feeding the same sample–recover–Lehmann pipeline, *not executed*

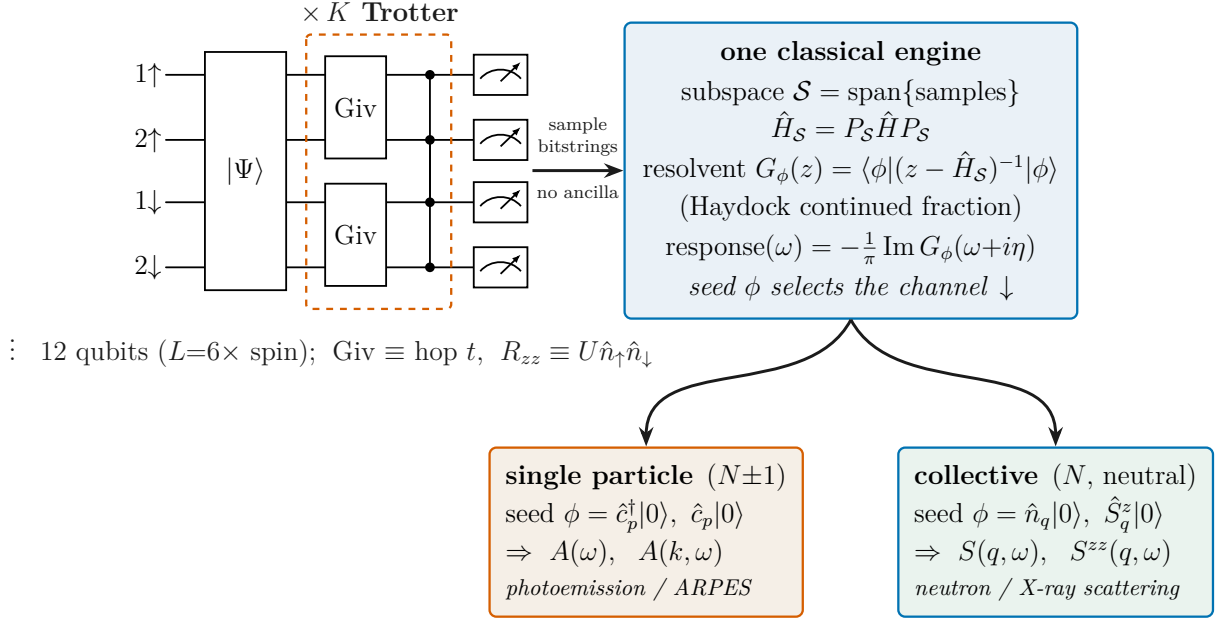


Figure 14. One sampling primitive, four dynamical responses. A shallow, ancilla-free circuit—state preparation and a Trotter layer repeated K times (Givens rotations for the hopping t , native R_{zz} for the on-site $U \hat{n}_{\uparrow} \hat{n}_{\downarrow}$)—is measured in the computational basis; no Hadamard test or controlled unitary is run on the device. The sampled bitstrings define a sector-specific subspace $\mathcal{S}$ and feed a single classical engine—a Haydock continued-fraction resolvent [54] $G_{\phi}(z) = \langle \phi | (z - \hat{H}_{\mathcal{S}})^{-1} | \phi \rangle$, whose single-particle case ($\phi = \hat{c}_p^{\dagger} |0\rangle$) is the Green’s function of Eq. (2). The seed ϕ then selects the channel: particle addition/removal seeds $\hat{c}_p^{\dagger} |0\rangle, \hat{c}_p |0\rangle$ in the $(N \pm 1)$ sectors give the single-particle spectral functions $A(\omega)$ and $A(k, \omega)$ (photoemission/ARPES), while number-conserving density and spin seeds $\hat{n}_q |0\rangle, \hat{S}_q^z |0\rangle$ in the N sector give the dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$ (neutron/X-ray scattering)—all from *one* sampling primitive, the seed selecting the sector and hence the subspace. The four channels are realized in Figs. 3 ($A(\omega)$), 4 ($A(k, \omega)$), 6 ($S(q, \omega)$), and 7 ($S^{zz}(q, \omega)$).

on hardware here—is provided as an ancillary file, bridging the concept from the Hubbard model to chemistry and situating the method within the quantum-centric-supercomputing program that couples IBM Heron processors to the Fugaku supercomputer [7, 15] under error-mitigated observable estimation [38, 110].

To take the same sample-recover pipeline to a molecular energy, we ran the ground state of stretched N_2 ($R = 2.0$ Å, valence active space) on a second Heron processor, `ibm_marrakesh` (10,000 shots), through a double-factorized Trotter circuit. Self-consistent configuration recovery reaches, averaged over 8 recovery seeds on the same device counts, $E_{\text{hw}} = -108.80745 \pm 0.00014$ Ha— 0.59 ± 0.14 mHa of the exact FCI value ($E_{\text{FCI}} = -108.80804$ Ha; best seed 0.40 mHa)—well below chemical accuracy on a real device for a strongly multireference bond (Fig. 15, panel b). Notably, the same shallow circuit evaluated by *noiseless* statevector simulation lands at -108.77850 Ha (29.5 mHa error): here the device noise, filtered through recovery, populates valid configurations the ideal shallow circuit misses, an instance of the noise-assisted recovery mechanism we analyze at length in a companion study [27]. The same effect has been reported independently on a variable-length cuprate chain, where real trapped-ion noise *improves* the sample-based energy convergence over the noiseless statevector simulation [119], and—on N_2 with the LUCJ ansatz, the present system—the same noise-enhanced configuration recovery is reported independently in [63], so the mechanism is not an artifact of one processor or one molecule. The quoted ± 0.14 mHa is the spread over the 8 recovery seeds—the stochastic uncertainty of the self-consistent recovery on the fixed device counts, not a device-repeat statistic; the systematic version, across nineteen molecules and a controlled noise model, is the subject of Fig. 17 below.

Why N_2 alone is not the test: noise robustness across chemistry. A single molecule cannot separate a genuine mechanism from a lucky geometry, so we stress-test the sample-recover energy pipeline across the entire nineteen-molecule suite of Table 1 under a controlled, hardware-motivated noise model. For each dissociated molecule we Born-sample the exact ground-state configuration distribution, apply an independent per-qubit bit-flip channel at rate ε , and rebuild the subspace either by naive in-sector post-selection or by self-consistent configuration recovery (S-CoRe) to the correct (N_α, N_β) sector; the recovered configurations are diagonalized and the energy is compared to FCI, seed-averaged (Fig. 17). Two facts emerge, and both bound the limits. First, at a realistic Heron-scale rate $\varepsilon = 2\%$ the subspace energy is intrinsically forgiving—because diagonalization in the sampled subspace is variational, corrupted bitstrings that survive as valid configurations only enlarge the subspace, a valid-shot economy rather than a fidelity gain—so 18/19 molecules reach chemical accuracy with recovery (the lone exception is the strongly multireference H_6 chain at 3.6 mHa), and S-CoRe converts one of the two naive failures (NH_3 , $2.65 \rightarrow 1.51$ mHa) into a success. Second, and more informative, the *advantage* of recovery grows monotonically with the noise: where naive post-selection discards the increasing fraction of out-of-sector strings and its error climbs steeply (N_2 : $0.4 \rightarrow 6.4 \rightarrow 31.9$ mHa at $\varepsilon = 0, 20, 30\%$), S-CoRe reassigns them to physical configurations and roughly halves the error throughout (N_2 : $0.4 \rightarrow 3.3 \rightarrow 13.2$ mHa; CO actually *improves*, from 1.7 mHa at $\varepsilon = 0$ to a 0.6 mHa minimum near $\varepsilon = 20\%$ (0.7 mHa at $\varepsilon = 30\%$), as recovered strings enrich the subspace). The effect is molecule-dependent: species whose ground state concentrates on few determinants (C_2) are essentially noise-immune out to $\varepsilon = 30\%$ (H_2O is likewise noise-immune at the realistic $\varepsilon = 2\%$), while the hardest (H_6) remains the residual challenge—consistent with the resource analysis of Sec. 3, where the cost is the determinant support $|\mathcal{S}|$ —tracked by the entanglement, not predicted by one-body magic. This is the molecular counterpart of the Hubbard noise sweep of Fig. 16, and it answers the natural objection to a single-molecule hardware run: the recovery mechanism is a property of the chemistry, not of N_2 .

Data and code availability. Both hardware runs are archived with full metadata. For each run we provide, as ancillary files, the raw measurement counts, the transpiled circuits (OpenQASM), the backend, calibration date, physical-qubit layout and coupling map, transpiled two-qubit-gate count and depth, job identifiers, the reference-state (LUCJ/Slater) parameters, the Trotter step and the real-time-evolution times $t_k = k \Delta t$ ($k = 0, \dots, 7$: the $t = 0$ reference plus the $K = 7$ evolutions) that define the union subspace, and the full mitigation settings (TREX, Pauli twirling, dynamical decoupling). The scripts that regenerate every figure and Table 1—including the per-molecule geometries, active spaces, FCI energies, absolute $|\mathcal{S}|$, sector dimension D , and minimal bond dimension χ of App. C—are provided alongside; every non-hardware result is an exact classical simulation reproducible from them. The complete reproduction repository—source, data, figure scripts, and a master reproduction notebook—is openly available at <https://github.com/nicolasbonilla/dynamical-spectral-functions-sqd>.

6. Discussion

6.1 An honest account of the classical hardness

We state plainly what is certified and what is open. Sampling from a subclass of unitary-cluster-Jastrow circuits is classically hard—it embeds instantaneous-quantum-polynomial computation [84, 85]—and adding a single non-Clifford resource already makes distribution learning hard [86], just as the difficulty of learning fermionic states grows with their non-Gaussian gate count [80]; the cost of matchgate simulation grows with the non-Gaussian (controlled-phase) content of the circuit [21]. These are not, however, a monotone certificate of hardness for a specific trained instance: the fermionic AntiFlatness is a *state* monotone while the matchgate simulation cost is a *circuit* quantity, and their relation is empirical rather than a theorem. A tensor-network comparison sharpens the scope. On one-dimensional chains a matrix-product state of small bond dimension represents these states far

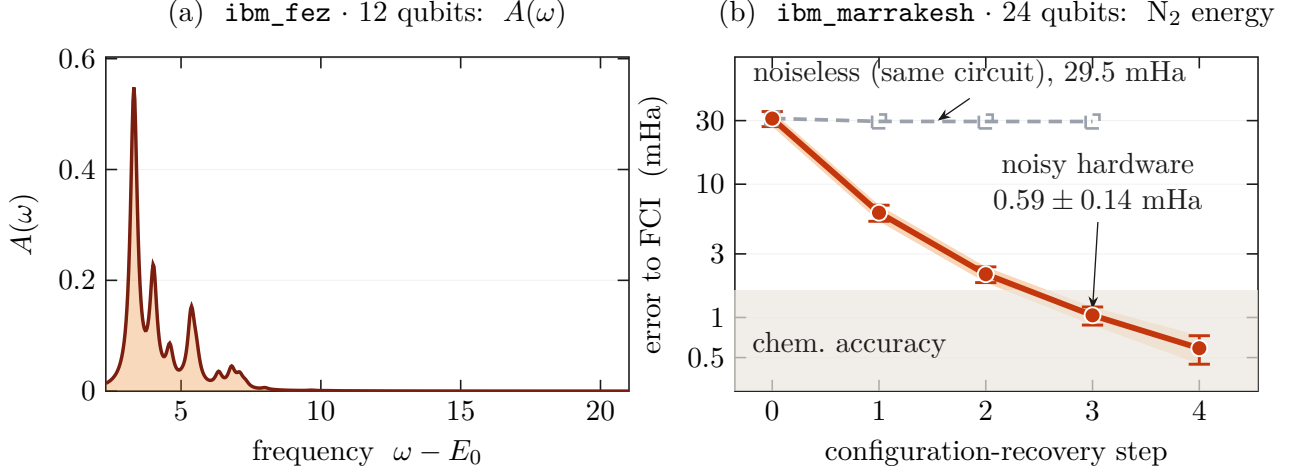


Figure 15. The complete quantum-hardware record of this work: two runs on IBM Heron processors. (a) Single-particle spectral function $A(\omega)$ of the $L = 6$ Hubbard model at $U/t = 4$, $\eta = 0.15t$ (12 qubits; the classical validation of Fig. 3 is a different instance, $U/t = 8$), reconstructed from 50,000 computational-basis bitstrings sampled on `ibm_fez` (TREX readout twirling, Pauli twirling, and dynamical decoupling). The device histogram covered the complete $(N+1)$ sector ($|\mathcal{S}| = 300/300$), so the reconstruction is exact by coverage—a proof of principle, not a fidelity or beyond-classical claim. (b) Ground-state energy error of stretched N_2 ($R = 2.0$ Å, CAS(10e,12o)/cc-pVDZ, 24 qubits) versus self-consistent configuration-recovery step, on `ibm_marrakesh` (10,000 shots, double-factorized Trotter circuit). The noisy device (solid) reaches 0.59 ± 0.14 mHa of the exact FCI value—well below chemical accuracy—whereas the noiseless statevector simulation of the *same* shallow circuit (dashed) is stranded at 29.5 mHa: device noise, filtered through configuration recovery, populates valid configurations the ideal shallow circuit misses. The solid curve is the mean over 8 recovery seeds on the cached device counts; the shaded envelope and the capped whiskers at every step both show the ± 1 s.d. spread across those 8 seeds (widest at step 0, 31.1 ± 4.0 mHa, where it starts together with the noiseless curve, and tapering as recovery converges to 0.59 ± 0.14 mHa; best seed 0.40 mHa). The systematic version across nineteen molecules under a controlled noise model is Fig. 17. These two runs are the entirety of the quantum hardware used here; every other result in this work is an exact classical simulation.

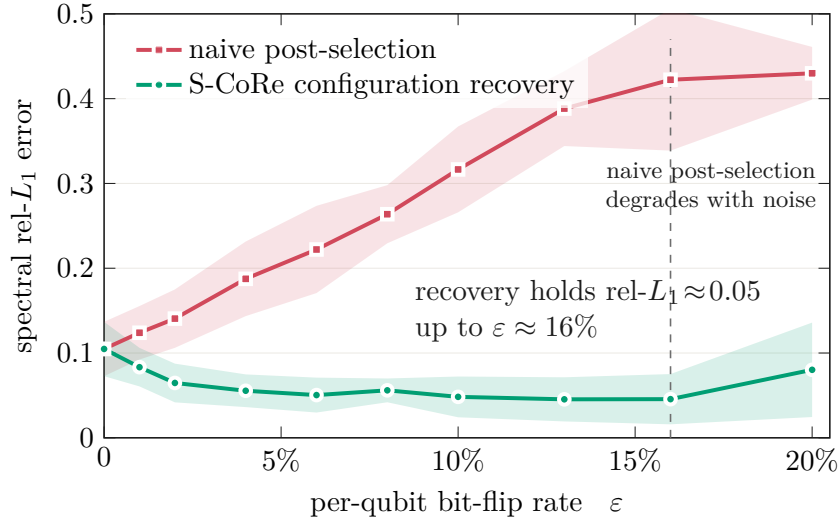


Figure 16. Noise robustness (synthetic per-qubit bit-flip channel, $L = 6$). Spectral relative- L_1 error vs the per-qubit bit-flip rate ϵ , applied to bitstrings Born-sampled from the exact distribution (this is a controlled classical simulation, not a device run). Naive post-selection degrades steadily, whereas S-CoRe self-consistent configuration recovery holds the error near 0.05 out to $\epsilon \approx 16\%$ (seed-averaged over 6 seeds at 6,000 shots each; bands = ± 1 s.d. across seeds).

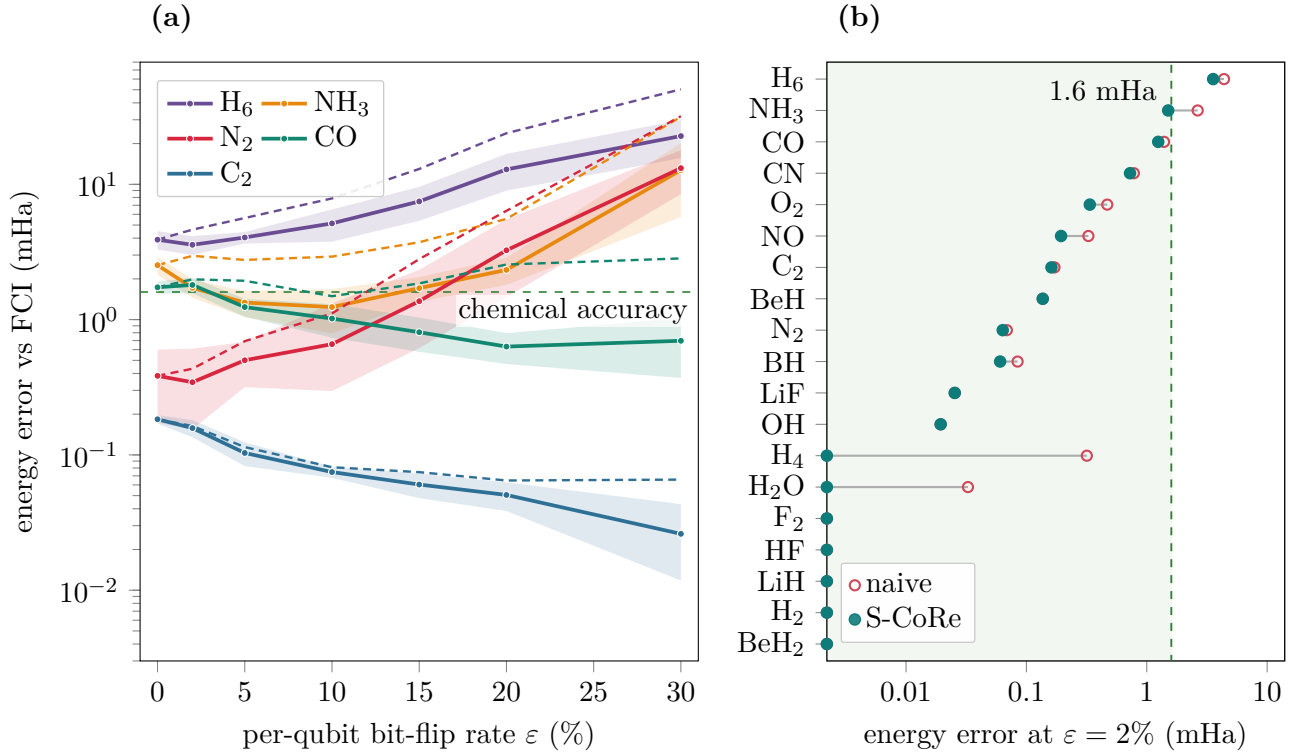


Figure 17. Noise-assisted SQD ground-state energy across the molecular suite. (a) Energy error vs the per-qubit bit-flip rate ε for five representative molecules (H_6 , NH_3 , N_2 , CO , C_2 ; seed-averaged over 8 seeds at 4,000 shots each, bands = ± 1 s.d. across seeds; log scale). Solid = S-CoRe recovery, dashed = naive post-selection. The recovery advantage grows with noise, and for CO and C_2 the recovered subspace drives the error *below* the noiseless value; C_2 is effectively noise-immune across the full ε sweep. (b) The full nineteen-molecule suite at a realistic $\varepsilon = 2\%$ (log scale, sorted): for each molecule the naive post-selection error (open circle) is linked to the S-CoRe value (filled), so the leftward pull is the recovery gain; the shaded band is chemical accuracy (< 1.6 mHa). Eighteen of nineteen land inside it; H_6 , the most multireference case, is the residual challenge. Panels (a) and (b) are independent seed ensembles (panel (a): 8 seeds; panel (b): 5 seeds, 4000 shots), so a given molecule’s $\varepsilon = 2\%$ value can differ by up to a factor of a few between them (for N_2 , 0.34 vs 0.06 mHa; both far below chemical accuracy).

more compactly than any determinant subspace—unsurprising, since one dimension is where the density-matrix renormalization group is near-optimal [131]. DMRG has matured into a production *ab initio* method for quantum chemistry [97, 98], and mature correction-vector, dynamical-DMRG, time-dependent-DMRG [83], and kernel-polynomial spectral-function methods already exist [88–93]; beyond tensor networks, classical Pauli-propagation methods now evaluate real-time correlators directly [82]. *We therefore do not claim advantage over tensor networks in one dimension.* The configuration-selection advantage we demonstrate—over deterministic Krylov and CIPSI selectors, not over heat-bath CI—is relevant in the settings where matrix-product representations strain: genuinely two-dimensional geometries and quantum chemistry, whose orbital graphs lack the one-dimensional locality that matrix-product states exploit. Figure 13 isolates the resource mechanism in a controlled contrast: at *matched* one-body magic a two-leg Hubbard ladder already carries a larger minimal bond dimension—and a $|\mathcal{S}|$ up to $\sim 2.2\times$ that of the chain, needing a correspondingly larger $(N\pm 1)$ fraction—the chain-to-ladder ratio growing with size. A ladder of this width ($\chi \approx 38$) remains classically near-exact for dynamical DMRG, so this is a resource contrast—the cost tracks χ , not $\mathcal{F}_1$, exactly as the resource theory predicts—not a beyond-classical claim. On the positive side, Theorem 1 (App. A) guarantees that a subspace containing the order- K Krylov space reproduces the exact spectral moments through order $2K+1$ and is captured with a polynomial shot budget, so the sampled reconstruction is convergent and its error is bounded by the captured spectral weight.

The strongest classical counterexamples, and why they sharpen rather than sink the thesis. Two recent results set the bar for any ground-state advantage claim, and we concede both without reservation. Reinholdt *et al.* [25] show that quantum-selected configuration interaction re-samples already-seen determinants in a coupon-collector process and returns expansions *less* compact than classical heat-bath or adaptive selected CI; Zhang and Otten [118] reach the ground state of the [4Fe-4S] cluster from *random* Slater determinants using of order 10^6 times fewer determinants and CPU-hours than the quantum demonstration. For the ground-state *energy* of the systems shown to date, strong classical selectors match or beat sample-based diagonalization, usually far more cheaply—we do not dispute it. Our decoupling explains *why* this must be so rather than contradicting it. The quantity these critiques probe is the size and compactness of the determinant expansion $|\mathcal{S}|$, a basis-dependent count that a free orbital rotation moves across its whole range and that the invariant one-body magic $\mathcal{F}_1$ is provably blind to (Sec. 3, Thm. 2). The critiques therefore land where our analysis predicts a classical method should win—on a basis-dependent, higher-order cost that no efficiently measurable state monotone we know can rank—and they say nothing about the invariant multireference diagnostic, which was never a cost claim. The conclusion is not that the critiques are wrong but that the advantage question has been posed about the wrong observable: the ground-state energy, where $|\mathcal{S}|$ is classically tamable and random- or heat-bath-selected determinants suffice, rather than the frequency-resolved response, whose faithful reconstruction demands a large fraction of the $(N \pm 1)$ sector (Sec. 3) and where the compactness edge that powers heat-bath and adaptive selected CI [94–96] does not transfer. We make no beyond-classical claim even there; we relocate the open question to where it remains open.

6.2 Machine learning that preserves, rather than removes, the quantum sampling role

Machine learning can enhance the pipeline only where it *consumes device samples or schedules device work*—the regime where learning from quantum experiments carries a provable advantage [103]—never where it could generate the dominant determinants unaided, which would dequantize the method exactly as the classical selectors of the introduction do [22, 23, 27]. We put the strongest candidate to a stringent test. A generative tail-completer—an orbital-native generative flow network [104] trained to sample high-weight $(N+1)$ -sector configurations—is asked to complete an under-sampled device head, in both device-seeded and from-scratch conditions (Fig. 18).

The result that carries the effect size here is not the generator but the baseline it fails to clear, and we state it in its own right rather than as a foil. Adding a cheap classical Epstein–Nesbet perturbative (CIPSI) tail [56] to the *same* device histogram—no additional shots, no additional quantum processing, and a first-order selection score that requires a single application of H to the sampled wavefunction—cuts the energy error from 39.24 ± 3.95 to 26.98 ± 0.66 mHa (mean $\pm$ s.d. over five seeds), a 31% reduction and 3.1 s.d. of the pre-correction spread. That 12.26 mHa is three times the 4.04 mHa the generative model then contributes on top of it, and it costs no quantum resources at all; it should be the default post-processing step on any under-sampled sample-based head; a demonstration that omits it is comparing against a baseline that is too weak. Only against that corrected baseline is the generative question well posed—and there the learned generator’s point estimate edges lower still but not by a statistically resolved margin (the further 4.0 mHa fused-vs-control difference is 1.2 s.d. of the generator’s own five-seed spread, against 3.1 for the classical step; Fig. 18), and the residual error is closed by drawing more device samples, not by the generator. This is a *null result*: on this single instance a learned generator does not beat the cheap classical baseline that the quantum sampling already enables—it does not, on the evidence here, manufacture dominant determinants the sampling misses. The safe roles for machine learning are therefore strictly post-processing and scheduling—mitigation and certification of the sampled histogram feeding recovery [38, 110], a role in which generative flow networks are already being deployed for measurement optimization—both to group commuting operators into compatible measurement sets [106] and to optimize the discrete measurement basis [107]—and an active-learning schedule that places the evolution times where the spectral function is most uncertain, in the

spirit of Bayesian optimization for inverse problems in quantum reaction dynamics [108]—none of which replaces the quantum sampling. The dequantization guard reported here corroborates, from the spectral-function side, the classical-simulability frontier mapped for ground-state SQD in our companion review [27].

This delimits, rather than forecloses, the “quantum-for-AI” horizon, and bounds where it could lie. The 2026 generative frontier has two mirror faces: Born machines that are classically *trainable* yet quantum-hard to *sample* [126, 127], and sample-based diagonalization, which is quantum-*sampled* yet classically *diagonalized*. The defensible role for a learned model is the one that respects this duality—consuming the device histogram rather than replacing it [103, 123]—and the theme of the present programme, amortization, names the opening the current instance-by-instance generative selectors [105, 124, 125] leave open. Per-molecule generative configuration recovery can already improve a *single* target’s sampled sector when retrained for that molecule [105]; what our negative result and this programme target is different—a recovery model trained *once* across a family to complete the physical-sector tail of a new, under-sampled instance without per-instance retraining. We take a first, falsifiable step on this question (Fig. 19). Trained once across a six-instance Hubbard family and evaluated leave-one-out, an amortized configuration-importance model recovers a held-out instance’s spectral support *at the level of* the strong per-instance classical selector (an Epstein–Nesbet tail it never sees): the leave-one-out relative- L_1 difference is -0.01 to -0.09 across subspace fractions, within ~ 1 s.d. over the six folds. Amortization without per-instance retraining is therefore *feasible*—a train-once model generalizes across the family—but shows *no statistically resolved advantage* over the classical baseline yet, the status of the open question: the single-instance negative result sets the bar, the amortized result meets but does not clear it, and—consistent with both—any advantage, if it exists, would be confined to the high-noise, high-correlation regime, never on the easy instances where the classical prior already wins.

6.3 Significance for academia and industry

Academia. The result reframes the quantum-advantage question in electronic structure. The debate has centred on the ground-state *energy*, where the evidence for a generic exponential advantage is weak [20] and the flagship sample-based demonstrations are being dequantized [22, 25]. We move the question to a regime where an advantage is defensible—dynamical response—and, more consequentially, replace an asserted advantage with a *precise resource statement*. One-body fermionic magic, though efficiently computable, is decoupled from the sampling cost and cannot certify where a quantum processor is required; what it faithfully reports is the static, multireference character of a target, while the operative hardness lives in the higher-order, connected-cumulant sector. In drawing this line we tie the resource theory of fermionic non-Gaussianity [68, 71, 76] to sample-based electronic-structure computation [7, 8] for the first time, and open a concrete programme: the frequency-resolved observables now accessible from sampled subspaces are the very objects of dynamical mean-field theory [101] and correlated spectroscopy [102], and the resource analysis is a template for how quantum-advantage claims can be *scoped and delimited* rather than asserted—a discipline the field increasingly demands [6, 20, 31].

Industry. The significance for the quantum-computing industry is threefold and concrete. *First, the right observables.* What the method produces—the single-particle spectral function measured by photoemission, its momentum-resolved form, and the dynamical structure factor probed by neutron and X-ray scattering [58, 59]—are precisely the quantities that materials discovery, catalysis, and correlated-electron device design require, and that the energy-only output of today’s sample-based hardware does not deliver [7, 11]. *Second, a correlation diagnostic.* Quantum runtime on the quantum-centric supercomputers now in operation—an IBM Heron processor coupled to the Fugaku supercomputer [7, 15], the QSCI software stack [8, 9], and the trapped-ion Green’s-function and dynamical-structure-factor pipeline [39, 49]—is scarce and expensive, so knowing *which* targets are strongly correlated is operationally valuable. The efficiently measurable fermionic magic flags exactly

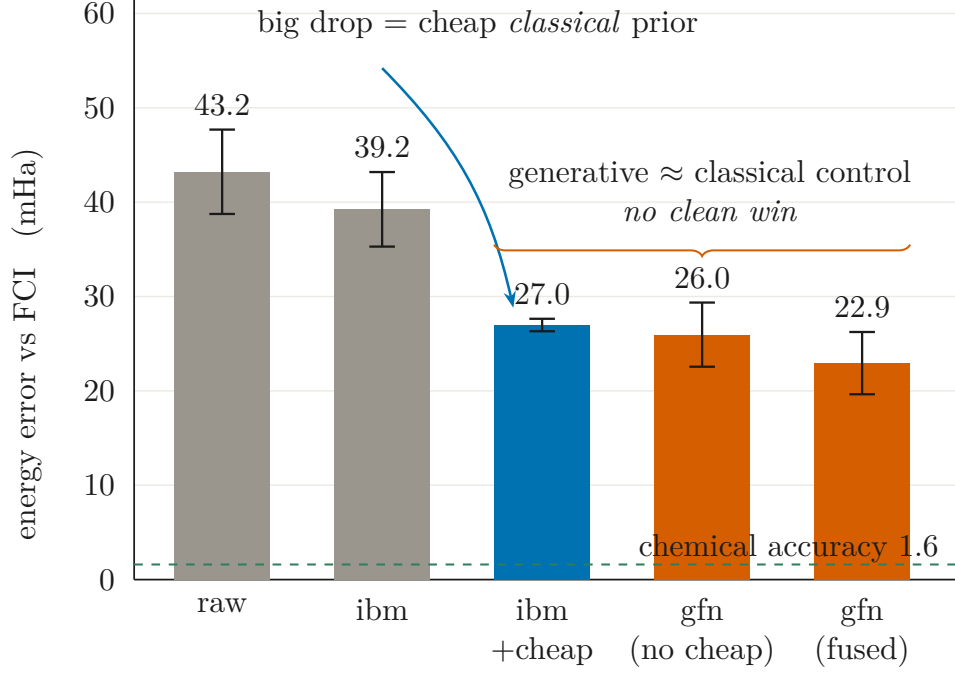


Figure 18. The large effect is the free classical tail; the learned generator adds no resolved gain. At an under-sampled device head (N_2 , 1000 shots, 5 seeds), a cheap classical Epstein–Nesbet (CIPSI) tail [56], drawing no additional shots, cuts the error from 39.24 ± 3.95 to 26.98 ± 0.66 mHa—a 31% reduction and 3.1 s.d. of the pre-correction spread. A generative GFlowNet tail-completer then edges the point estimate lower still (25.96 ± 3.40 mHa from scratch, 22.94 ± 3.30 mHa fused onto the classical prior), but that further 4.0 mHa fused-vs-control improvement—a third the size of the classical step—is only 1.2 s.d. over five seeds, not a statistically resolved win. The residual error is closed by drawing more device samples, not by the generator: on this single instance the quantum sampling, not a learned generator, supplies the dominant determinants that additional device shots recover.

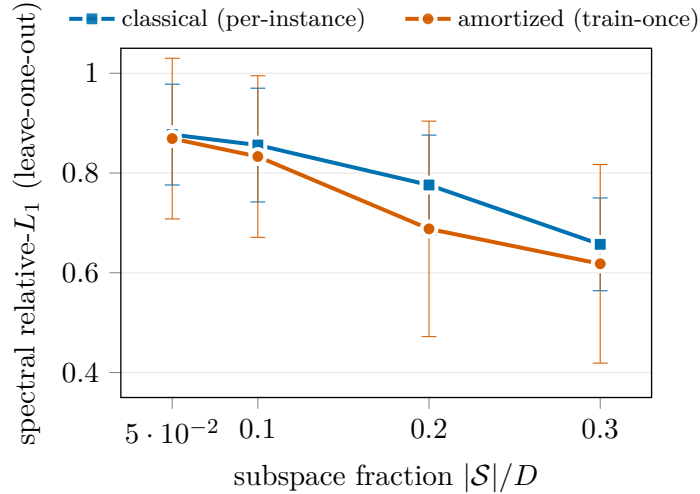


Figure 19. Amortized configuration recovery: a first data point on the programme’s open question. A configuration-importance model trained *once* across a six-instance Hubbard family ($L = 6, 8$; $U/t = 4, 8, 12$) and evaluated *leave-one-out* recovers a held-out instance’s $(N+1)$ -sector spectral support, versus the strong per-instance classical selector (Epstein–Nesbet, untrained), as a function of the sampled subspace fraction. The amortized model—which never sees the held-out instance—meets the classical baseline (leave-one-out $\Delta = \text{amortized} - \text{classical} = -0.01$ to -0.09 across fractions, within ~ 1 s.d. over the six folds): amortization without per-instance retraining is *feasible* but shows no statistically resolved advantage yet. Error bars are ± 1 s.d. across the six leave-one-out folds.

the multireference regime where mean-field and weakly-correlated classical methods break down. It is not, however, a cost oracle: being decoupled from the determinant support (Sec. 3), it cannot by itself rank a strongly-correlated target against a classical selected configuration interaction [95, 96], the density-matrix renormalization group [87], or auxiliary-field quantum Monte Carlo [99]—that decision turns on the higher-order, basis-dependent cost the one-body magic does not see [25]. *Third, low adoption cost.* The method is hardware-native: it needs only computational-basis sampling, with no Hadamard tests or controlled unitaries [47, 48], so it runs on current devices within the existing SQD stack and its GPU-accelerated classical back end [121, 122], and it *lightens* rather than adds to the observable-estimation and error-mitigation burden that dominates near-term cost [37, 38, 110]. Fermionic magic is thus offered as an efficiently measurable multireference diagnostic—a *one-sided* triage that flags, before committing runtime, the strongly-correlated targets on which mean-field and weakly-correlated classical methods fail. It is expressly *not* a placement oracle: whether a workload belongs on the quantum processor turns on the basis-dependent cost—the bond dimension χ —that one-body magic provably cannot supply (Sec. 3), and we claim no scheduling result absent a scheduling experiment.

7. Conclusion

The pressing question for sample-based quantum computation is no longer whether it can match classical methods on the ground-state energy—the classical frontier has answered that in the negative [20, 22, 25]—but what its output should be, and what resource actually sets its cost. This work gives a two-part answer. We introduced a method that computes dynamical spectral functions $A(\omega)$, $A(k, \omega)$, and $S(q, \omega)$ from purely bitstring-sampled quantum subspaces, lifting the SQD/QSCI paradigm from static energies to frequency-resolved dynamics with no Hadamard tests, validated against exact diagonalization across Hubbard chains and nineteen molecules and demonstrated on the IBM Heron processor—the $L = 6$ Hubbard spectral function $A(\omega)$ reconstructed on `ibm_fez`, and, for the ground-state energy of stretched N_2 , a noise-assisted configuration-recovery run on `ibm_marrakesh` reaching 0.59 ± 0.14 mHa of the exact value (over 8 recovery seeds).

We then asked which resource controls the determinant support $|\mathcal{S}|$ that the sampler must populate, and corrected a natural expectation. Restricted to the number-conserving states of chemistry, the fermionic AntiFlatness collapses to a single one-particle-density-matrix invariant, $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ (equal to $2N_u$ for spin-restricted natural orbitals), identifying fermionic magic with the effectively-unpaired-electron index and one-body entanglement. Being a Gaussian (orbital-rotation) invariant while $|\mathcal{S}|$ is basis dependent, it is provably decoupled from the sampling cost—two-sidedly and without a stable sign—and no orbital-rotation-invariant functional of the reduced density matrices can repair this. Across the nineteen-molecule suite one-body magic is nonetheless a faithful, efficiently measurable detector of static, multireference character—switching on for covalent bond-breaking, already large for inherently multireference C_2 and O_2 , near-Gaussian for the ionic bond of LiF —but it is a diagnostic of correlation, not a predictor of classical cost—which is instead lower-bounded and, on the systems realized here, empirically tracked by the basis-dependent entanglement, the bond dimension χ . Any genuine quantum advantage lives one stratum above, in the generic, high-rank non-Gaussianity of the connected higher-body cumulants in the fixed sampling basis—not any orbital-invariant magic order—beyond the stratum whose free-fermionic estimation primitives remain classically tractable [77].

We have been deliberate about scope. The reach of the method is asymptotic and lives in two-dimensional and chemical settings rather than one dimension, where tensor networks remain near-optimal [131]; it rests on a moment-exactness and polynomial-shot theorem (Thm. 1, App. A); and, on the artificial-intelligence side of the hybrid loop, self-consistent configuration recovery improves the sampled subspace under realistic device noise through valid-shot economy rather than quantum advantage, the large effect in the dequantization test is a free classical one—an Epstein–Nesbet tail on the existing device histogram, no extra shots, cutting the energy error by 31% ($39.24 \pm 3.95 \rightarrow 26.98 \pm 0.66$ mHa)—and a *learned* generative tail-completer does *not* beat that

corrected baseline (Fig. 18)—delimiting, not foreclosing, the quantum–AI frontier, and consistent with the classical-simulability boundary. Read together, the two contributions deliver a dynamical primitive the energy-only hardware of today does not provide, and a precise account of which fermionic resource sets its cost and which merely diagnoses correlation—handing the quantum-centric supercomputing programs now operating at industrial scale [7, 15] a computable multireference diagnostic—and a proof that this easily-measured quantity does not by itself map the classical cost frontier, which is instead lower-bounded and empirically tracked by the bond dimension χ . The natural next steps are to push the sampled-subspace reconstruction into the two-dimensional and multi-orbital regimes where the higher-order hardness lives, and to make the *basis-dependent* cost—the bond dimension χ in the sampler’s own frame, and the generic high-rank non-Gaussianity above it—not the orbital-invariant one-body magic that is so easily measured—the object one estimates when deciding whether a quantum sampler is worth its shots.

Appendix A. Moment-exactness and a sample-complexity bound

We make the convergence of the bitstring-sampled spectral function precise. Fix the seed $|\phi\rangle = c_p^\dagger |\Psi_0\rangle$ (the add-electron branch; the removal branch is identical), with weight $\|\phi\|^2 = \langle\phi|\phi\rangle$, and let $A(\omega) = \sum_n |\langle n|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (E_n - E_0))$ be the exact spectral function, where $\{|n\rangle, E_n\}$ diagonalize H in the $(N+1)$ -particle sector and $\mathcal{L}_\eta(x) = \frac{\eta/\pi}{x^2 + \eta^2}$. For a sampled configuration set S let P_S project onto $\text{span}(S)$, let $H_S = P_S H P_S$ with eigenpairs $\{|\tilde{m}\rangle, \tilde{E}_m\}$, and let $A_S(\omega) = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \mathcal{L}_\eta(\omega - (\tilde{E}_m - E_0))$ be the reconstruction. Write $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K\phi\}$ for the order- K Krylov space of the seed and $w_S = \|P_S\phi\|^2 / \|\phi\|^2$ for the captured spectral weight.

Theorem 1 (Moment exactness, sampling capture, and error bound). (i) Moment exactness. *Let $d\mu(\omega) = \sum_n |\langle n|\phi\rangle|^2 \delta(\omega - (E_n - E_0))$ and $d\mu_S(\omega) = \sum_m |\langle \tilde{m}|\phi\rangle|^2 \delta(\omega - (\tilde{E}_m - E_0))$ be the unbroadened ($\eta \rightarrow 0$) Stieltjes measures whose Lorentzian convolutions are A and A_S . If $\text{span}(S) \supseteq \mathcal{K}_K$, then A_S reproduces the exact spectral moments through order $2K + 1$,*

$$\int \omega^j d\mu_S(\omega) = \int \omega^j d\mu(\omega) = \langle\phi|(H - E_0)^j|\phi\rangle, \quad 0 \leq j \leq 2K + 1. \quad (5)$$

The moments are taken on the point measure $d\mu$, not on the Lorentzian-broadened A (whose moments of order ≥ 2 diverge). Consequently, at fixed broadening $\eta > 0$, the smoothed reconstruction converges geometrically, $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$, where both the prefactor $C(\eta) \rightarrow \infty$ and the rate $\rho(\eta) \rightarrow 1^+$ as $\eta \rightarrow 0^+$ —the resolution/convergence trade-off of Gauss–Lanczos quadrature on a Lorentzian-smoothed measure.

(ii) Sampling capture. *Draw T bitstrings from the time-averaged Born mixture $p(x) = \frac{1}{K+1} \sum_{k=0}^K |\langle x|e^{-iHt_k}\phi\rangle|^2 / \|\phi\|^2$. Every configuration with $p(x) \geq p_{\min}$ is included in S with probability at least $1 - (1 - p_{\min})^T$; hence capturing all configurations carrying mixture-weight $\geq \varepsilon$ with confidence $1 - \delta$ requires only*

$$T = O(\varepsilon^{-1} \log(|S|/\delta)) \quad (6)$$

shots — polynomial in the target subspace size and independent of the Hilbert-space dimension.

(iii) Error bound. *For any S ,*

$$\|A - A_S\|_1 \leq 2 \|\phi\|^2 (1 - w_S) + \underbrace{C \rho^{-K_S}}_{\text{in-subspace quadrature}}, \quad (7)$$

where K_S is the largest Krylov order contained in $\text{span}(S)$.

Proof sketch. (i) The nonorthogonal Lanczos/Krylov construction makes A_S a Gaussian-quadrature approximation of the Stieltjes measure $d\mu(\omega) = \sum_n |\langle n|\phi\rangle|^2 \delta(\omega - (E_n - E_0))$; an order- $(K+1)$ quadrature matches the first $2K + 1$ moments $\mu_j = \langle\phi|(H - E_0)^j|\phi\rangle$ exactly, and the quadrature error on a smooth test function decays geometrically with the number of matched moments for a

measure of bounded support (the classical Gauss–Lanczos quadrature convergence result [53]). (ii) is a union bound over configurations of the coupon-collector event “ x never sampled in T draws,” whose probability is $(1 - p(x))^T \leq e^{-p(x)T}$; setting $e^{-\varepsilon T} \leq \delta/|S|$ gives the stated T . (iii) Split $\phi = P_S\phi + (\mathbb{I} - P_S)\phi$: the discarded component contributes total spectral weight $\|(\mathbb{I} - P_S)\phi\|^2 = \|\phi\|^2(1 - w_S)$, and since A and A_S are each nonnegative with $\int A = \|\phi\|^2$, $\int A_S = \|P_S\phi\|^2$, the L_1 discrepancy of the missing mass is bounded by twice that weight; the retained mass reproduces A up to the order- K_S quadrature error of part (i). $\square$

The three parts compose into the operational statement of the paper: the quantum computer supplies, in $\text{poly}(|S|)$ shots, a subspace whose captured weight $w_S \rightarrow 1$. This composition invokes the empirically observed residue concentration of Sec. 2 (that the high-Born-weight configurations coincide with the Krylov-spanning, high-seed-overlap set): absent it, parts (i) and (iii) certify convergence *given* a captured Krylov-adequate subspace, but not that $\text{poly}(|S|)$ sampling *finds* it. Whether that subspace is then *small*—whether $|S|$ is affordable—is set by the higher-order structure of the moments $\mu_j = \langle \phi | (H - E_0)^j | \phi \rangle$ (Sec. 3), a basis-dependent quantity that the one-body magic $\mathcal{F}_1$ provably cannot see.

Appendix B. One-body magic cannot bound the determinant support

Proposition 1 (Coefficient-insensitivity obstruction). *Let μ be any functional of a fixed finite set of reduced density matrices, continuous in the state. Then μ cannot lower-bound the raw determinant support (the unthresholded Slater rank); like the CP tensor rank, this count is not lower-semicontinuous [117]: for every $c > 0$ there exist states with μ arbitrarily small and rank arbitrarily large.*

Proof. Rank is a coefficient-insensitive count. Take $|\Psi_\epsilon\rangle = |D_0\rangle + \epsilon \sum_{k=1}^K |D_k\rangle$ with the D_k distinct. As $\epsilon \rightarrow 0$ every reduced density matrix converges to that of $|D_0\rangle$, so any continuous $\mu \rightarrow \mu(|D_0\rangle)$ (for $\mathcal{F}_1$, to 0), while the rank remains $\Theta(K)$. $\square$

The obstruction is about the unthresholded rank; the operative, coefficient-*sensitive* support $|S|_\epsilon$ of Sec. 3 collapses to 1 along the same $\epsilon \rightarrow 0$ path, so continuity alone does not obstruct it. What obstructs $|S|_\epsilon$ itself is the orbital-rotation symmetry—no invariant functional survives it—together with the two equal-weight witnesses below, whose amplitudes are $\Theta(1)$ and therefore threshold-robust. Two explicit witnesses rule out the natural repairs. The two-determinant cat $(|1 \cdots 10 \cdots 0\rangle + |0 \cdots 01 \cdots 1\rangle)/\sqrt{2}$ has $\gamma = \frac{1}{2}\mathbb{I}$, hence $\mathcal{F}_1 = 2n_o$ (maximal) yet Slater rank 2, so no $\mathcal{F}_1$ -monotone bounds $|S|$ from below; a paired (BCS) Gaussian state has Gaussian rank 1 yet fractional occupations, so no occupation functional bounds it from above. The only computable lower bounds on $|S|$ are the basis-dependent orbital-Schmidt (matrix-flattening) ranks [131]—the tensor-network bond dimension—to which any Gaussian-invariant magic measure is blind. That lower bound is exact:

Proposition 2 (The bond dimension lower-bounds the support). *For any state and any bipartition of the spin-orbitals into (L, R), the Schmidt rank χ across the cut (the minimal matrix-product-state bond dimension there) satisfies*

$$\chi \leq |S|, \quad (8)$$

where $|S|$ is the number of fixed-basis Slater determinants of nonzero amplitude. Hence the minimal bond dimension is a rigorous, basis-dependent lower bound on the determinant support. The bound is one-sided: the geminal witness of Thm. 2 has $\chi = 2$ yet $|S|_{\text{pair}} = 2^K$, so $|S|$ can exceed χ exponentially in a generic basis (and descends toward χ in the natural-orbital basis), while the orbital-invariant one-body magic $\mathcal{F}_1$ bounds neither (Prop. 1).

Proof. Write $|\Psi\rangle = \sum_{D \in \mathcal{S}} c_D |D\rangle$ over its support. Each determinant factorizes across the cut, $|D\rangle = |D_L\rangle \otimes |D_R\rangle$, so the matricization $A_{D_L, D_R} = c_D$ has at most $|\mathcal{S}|$ nonzero entries; its rank—the Schmidt rank χ —is therefore at most $|\mathcal{S}|$. $\square$

Since a matrix-product-state simulation costs $\text{poly}(\chi)$, Prop. 2 says that *in the sampler’s own basis a tensor network is never less compact than the sampled subspace*: this is why we make no beyond-classical claim in one dimension, where χ is small, and why any advantage must live in the higher-entanglement regime where χ itself—and with it the classical tensor-network cost—grows (the chain-to-ladder jump of Fig. 13). Empirically χ is also the operative correlate of the *measured* cost: across the 74 exact ground states of Fig. 2 the thresholded support $|\mathcal{S}|_\varepsilon$ tracks χ (pooled Spearman $\rho = 0.72$; $\rho = 0.90$ within the $n = 38$ molecular suite, where however one-body magic scores a comparable $\rho = 0.86$ —the two resources are separated not there but by the Hubbard sweep, $\rho = 0.57$ versus -0.11). Two caveats belong with that number. A rank correlation is not a statement of tightness, and per the geminal witness the bound can be exponentially loose. And the proposition is proved for the raw support: once amplitudes are thresholded the inequality need not survive—here $|\mathcal{S}|$ is cut at 99% of the weight while χ is cut at $\varepsilon = 0.05$, a tighter tolerance—and $|\mathcal{S}|_\varepsilon$ falls below χ in 7 of the 38 molecular points. What stands is that χ is a rigorous floor on the raw support and an empirical correlate of the measured one, whereas $\mathcal{F}_1$ is decoupled from both.

B.1 A provably-simulable witness: magic without cost

The obstruction of Prop. 1 is a continuity argument. We now sharpen it to a constructive, physically realizable family on which *every* order of fermionic magic is large yet classical simulation is provably efficient—and, being a member of a recently characterized simulable class, the witness is anchored to an external hardness result rather than to our own construction. Let $|\Psi_K\rangle$ be an antisymmetrized product of K strongly orthogonal singlet geminals (perfect-pairing / generalized-valence-bond),

$$|\Psi_K\rangle = \bigotimes_{i=1}^K \left(\cos \theta \hat{b}_{i\uparrow}^\dagger \hat{b}_{i\downarrow}^\dagger + \sin \theta \hat{a}_{i\uparrow}^\dagger \hat{a}_{i\downarrow}^\dagger \right) |0\rangle, \quad (9)$$

on $2K$ spatial orbitals ($4K$ spin-orbital modes, $N = 2K$ electrons), each geminal occupying a disjoint bonding/antibonding pair (b_i, a_i) .

Theorem 2 (Decoupling witness). *At maximal pairing $\theta = \pi/4$ the family (9) satisfies, exactly,*

$$\mathcal{F}_1 = 2N_u = 4K, \quad \mathcal{F}_2 = 4K, \quad |\mathcal{S}|_{\text{pair}} = 2^K, \quad \chi = 2, \quad (10)$$

where $\mathcal{F}_{1,2}$ are the order-1 and order-2 antifatness of Eq. (3), $|\mathcal{S}|_{\text{pair}}$ is the determinant support in the pairing basis, and χ is the minimal matrix-product-state bond dimension. Consequently one-body magic $\mathcal{F}_1$, higher-order magic $\mathcal{F}_2$, and determinant support $|\mathcal{S}|_{\text{pair}}$ all diverge with K while, in the localized (pairing) basis, the state remains a bond-dimension-2 tensor network—classically representable, contractible, and Born-samplable in $O(K)$: magic of every order is unbounded at fixed $O(1)$ simulation cost. (For this paired class the additive-error estimation of amplitudes, overlaps, and number correlators under free-fermionic dynamics is moreover classically efficient [77]; we invoke that result only for those primitives, and make no sampling claim about the rotated orbit discussed below.)

Proof. Each geminal is supported on its own four spin-orbitals, so (9) is a product state across the K blocks; its reduced one-particle density matrix is block diagonal with spatial natural occupations $(n_{b_i}, n_{a_i}) = (2 \cos^2 \theta, 2 \sin^2 \theta)$. Hence $N_u = \sum_a n_a (2 - n_a) = K[2 \cos^2 \theta (2 - 2 \cos^2 \theta) + 2 \sin^2 \theta (2 - 2 \sin^2 \theta)] = 8K \cos^2 \theta \sin^2 \theta$, giving $N_u = 2K$ and $\mathcal{F}_1 = 2N_u = 4K$ at $\theta = \pi/4$ by the collapse identity Eq. (4). The Majorana covariance is likewise block diagonal. At $\theta = \pi/4$ a single maximally-paired geminal is the two-determinant cat, whose covariance *vanishes identically* ($\Gamma = 0$): all its singular values are zero, so every order saturates its maximum,

$\mathcal{F}_k = 2n_o = 4$ per geminal for *all* k , giving $\mathcal{F}_2 = 4K$. This equality of orders is a knife-edge of maximal pairing; away from it the orders separate ($\theta = \pi/6$ gives $\mathcal{F}_1 = 3$, $\mathcal{F}_2 = 3.75$ per pair, both large on the same family). In the pairing basis each geminal is a two-configuration superposition, so the product carries 2^K determinants, $|\mathcal{S}|_{\text{pair}} = 2^K$. Finally, as a product of two-orbital factors the state is a matrix-product state whose every bond crosses at most one geminal, so $\chi = 2$ in this ordering; contraction and Born sampling then cost $O(K)$. All four values are reproduced to machine precision by exact diagonalization for $K = 1, \dots, 4$. $\square$

The witness (Fig. 20) is not the trivial “a product state is easy”; its force is in the *orbit*. Because $\mathcal{F}_1$ and $\mathcal{F}_2$ are invariant under number-conserving free-fermionic (orbital-rotation) unitaries U [71], the entire orbit $\{U|\Psi_K\rangle\}$ carries the *same* $(\mathcal{F}_1, \mathcal{F}_2) = (4K, 4K)$ while the classical sampling cost in the fixed computational basis—the bond dimension and Born-sampling time—ranges from $O(K)$ at the localized representative up to exponential (the minimal determinant support is $|\mathcal{S}|_{\text{pair}} = 2^K$ throughout). Two conclusions follow, each stated in the only direction the construction supports. (i) *A Gaussian-invariant fermionic magic is not a sufficient certificate of hardness and does not lower-bound the cost*: at the localized representative $\chi = 2$ (trivially easy) yet $\mathcal{F}_k = 4K \rightarrow \infty$; large magic confers no hardness. (ii) *$\mathcal{F}_k$ does not determine the cost*: it is constant on an orbit whose sampling cost varies from $O(K)$ to exponential, so the cost cannot be a function of it. Indeed in a generic rotated basis the computational-basis *sampling* of $|\Psi_K\rangle$ is plausibly classically hard—precisely the free-fermionic-linear-optics-with-magic-input regime of the Fermion Sampling advantage scheme [129]—while $\mathcal{F}_k$ is unchanged along the orbit: any hardness that appears is a property of the sampler’s fixed frame, invisible to the invariant magic. The construction is silent on whether magic is *necessary* for hardness (it may well be, and the paired stratum’s residual structure is where [77] locates tractability); it refutes only that magic is *sufficient* for, or *determines*, the cost—the coefficient-insensitivity of Prop. 1 made constructive. What does track the cost is the bond dimension in the operative sampling basis—not orbit-invariant. It *lower-bounds* the raw $|\mathcal{S}|$ and can be exponentially loose (this very witness has $\chi = 2$ while $|\mathcal{S}|_{\text{pair}} = 2^K$), yet across the realized molecular suite and Hubbard set it co-varies with the measured $|\mathcal{S}|_\epsilon$ (Spearman $\rho = 0.90/0.57$) and with the natural-orbital participation entropy that upper-bounds Gaussian-basis simulation cost [76]. Fermionic magic diagnoses correlation; entanglement in the sampler’s frame lower-bounds and tracks its price.

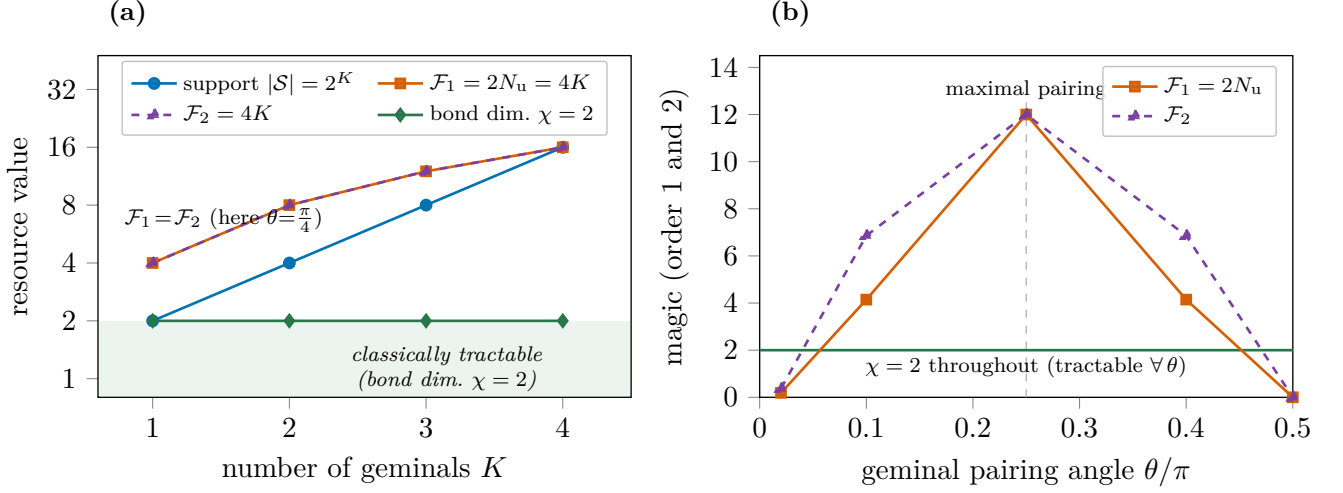


Figure 20. The decoupling witness (Theorem 2). Antisymmetrized products of strongly orthogonal geminals, shown in their localized (pairing) basis, where the minimal bond dimension is $\chi = 2$ and the state is a classically tractable tensor network. **(a)** As the number of geminals K grows, one-body magic $\mathcal{F}_1 = 2N_u = 4K$, higher-order magic $\mathcal{F}_2 = 4K$, and pairing-basis determinant support $|\mathcal{S}| = 2^K$ all diverge, yet the bond dimension never exceeds $\chi = 2$: the localized state never leaves the tractable region while magic of every order runs to infinity. **(b)** At $K = 3$, sweeping the pairing angle drives both $\mathcal{F}_1$ and $\mathcal{F}_2$ to their maximum at $\theta = \pi/4$ —where the covariance vanishes and all orders saturate—and $\mathcal{F}_2 > \mathcal{F}_1$ off the peak, while $\chi = 2$ throughout. Because $\mathcal{F}_k$ is orbital-rotation invariant, the same values persist on the whole orbit, along which the sampling cost ranges from $O(K)$ to exponential (Thm. 2): magic does not determine the price.

Appendix C. Reproducibility data for the molecular suite

Table 2 lists, for every molecule of Table 1, the equilibrium and dissociation bond lengths, the active space, the N -electron (neutral, ground-state) sector dimension D , the ground-state determinant support $|\mathcal{S}|$ (the number of determinants carrying $\geq 99\%$ of the FCI weight), and the minimal matrix-product-state bond dimension χ at both geometries. These are the quantities that enter the $\mathcal{F}_1$ - $|\mathcal{S}|$ and χ - $|\mathcal{S}|$ correlations of Sec. 3, pooled over the $n = 38$ equilibrium-and-dissociation points (Spearman χ - $|\mathcal{S}|$ $\rho = 0.90$, $\mathcal{F}_1$ - $|\mathcal{S}|$ $\rho = 0.86$; Pearson $\mathcal{F}_1$ - $\log_2 |\mathcal{S}|$ $r = 0.80$). The $|\mathcal{S}|/D$ reported in Table 1 is a *distinct* quantity—the fraction of the $(N+1)$ sector used in the spectral *reconstruction*, not the ground-state support tabulated here. Every active-space FCI energy is converged to better than 10^{-12} Ha (the largest residual across all 38 points is 1.1×10^{-13} Ha), comfortably below the 10^{-5} Ha quoted in the main text; the ground-state energies, full Cartesian geometries, and orbital active-space indices are provided as ancillary files.

Table 2. Reproducibility data for the nineteen-molecule suite (cc-pVDZ, valence bond-breaking active spaces). $R_{\text{eq}}, R_{\text{diss}}$ in Å; for the polyatomics ($\text{BeH}_2, \text{H}_2\text{O}, \text{NH}_3, \text{H}_4, \text{H}_6$) the representative varied bond length is shown and full Cartesians are in the ancillary data. D : dimension of the N -electron (neutral, ground-state) sector, in which the tabulated $|\mathcal{S}|$ lives; $|\mathcal{S}|$: ground-state determinant support ($\geq 99\%$ FCI weight); χ : minimal MPS bond dimension across the central spin-orbital cut. The $|\mathcal{S}|/D$ of Table 1 is a distinct ratio, taken in the $(N\pm 1)$ spectral sector.

Molecule	CAS(n_e, n_o)	R_{eq}	R_{diss}	D	$ \mathcal{S} ^{\text{eq}}$	$ \mathcal{S} ^{\text{diss}}$	χ^{eq}	χ^{diss}
F_2	(1+1,2)	1.412	3.00	4	1	2	2	2
N_2	(3+3,6)	1.098	2.40	400	7	80	7	32
H_2O	(2+2,4)	0.957	2.30	36	1	13	1	13
HF	(1+1,2)	0.917	2.30	4	1	4	1	4
NH_3	(3+3,6)	1.012	2.33	400	1	68	4	42
CO	(3+3,6)	1.128	2.40	400	8	46	14	30
H_6	(3+3,6)	0.900	2.00	400	9	91	8	36
H_4	(2+2,4)	0.900	2.00	36	2	11	4	8
H_2	(1+1,2)	0.741	2.00	4	1	2	1	2
BeH_2	(2+2,4)	1.334	2.90	36	1	6	1	8
C_2	(4+4,6)	1.243	2.60	225	13	6	11	2
NO	(4+3,6)	1.151	2.50	300	5	25	11	7
O_2	(5+3,6)	1.208	2.60	120	11	4	4	2
CN	(4+3,6)	1.172	2.50	300	27	9	17	7
OH	(3+2,4)	0.970	2.40	24	1	5	1	4
BH	(2+2,4)	1.232	2.60	36	5	3	2	2
BeH	(2+1,3)	1.343	2.70	9	1	2	1	2
LiH	(1+1,2)	1.596	3.20	4	1	2	1	4
LiF	(1+1,2)	1.564	3.20	4	1	1	1	1

References

- [1] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, J. L. O’Brien, “A variational eigenvalue solver on a quantum processor,” *Nat. Commun.* **5**, 4213 (2014); [arXiv:1304.3061](#).
- [2] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, J. M. Gambetta, “Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets,” *Nature* **549**, 242 (2017); [arXiv:1704.05018](#).
- [3] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven, “Barren plateaus in quantum neural network training landscapes,” *Nat. Commun.* **9**, 4812 (2018); [arXiv:1803.11173](#).
- [4] M. Cerezo *et al.*, “Variational quantum algorithms,” *Nat. Rev. Phys.* **3**, 625 (2021); [arXiv:2012.09265](#).
- [5] J. Tilly *et al.*, “The variational quantum eigensolver: a review of methods and best practices,” *Phys. Rep.* **986**, 1 (2022); [arXiv:2111.05176](#).
- [6] M. Cerezo, M. Larocca, D. García-Martín *et al.*, “Does provable absence of barren plateaus imply classical simulability?,” *Nat. Commun.* **16**, 7907 (2025); [arXiv:2312.09121](#).
- [7] J. Robledo-Moreno, M. Motta *et al.*, “Chemistry beyond the scale of exact diagonalization on a quantum-centric supercomputer,” *Sci. Adv.* **11**, eadu9991 (2025); [arXiv:2405.05068](#).
- [8] K. Kanno, M. Kohda, R. Imai, S. Koh, K. Mitarai, W. Mizukami, Y. O. Nakagawa, “Quantum-selected configuration interaction,” *Phys. Rev. Research* **8**, 023268 (2026); [arXiv:2302.11320](#).
- [9] Y. O. Nakagawa, M. Kamoshita, W. Mizukami, S. Sudo, Y. Ohnishi, “ADAPT-QSCI: adaptive construction of an input state for quantum-selected configuration interaction,” *J. Chem. Theory Comput.* **20**, 10817 (2024); [arXiv:2311.01105](#).

- [10] M. Mikkelsen, Y. O. Nakagawa, “Quantum-selected configuration interaction with time-evolved state,” *Phys. Rev. Research* **7**, 043043 (2025); [arXiv:2412.13839](#).
- [11] S. Barison, J. Robledo Moreno, M. Motta, “Quantum-centric computation of molecular excited states with extended sample-based quantum diagonalization,” *Quantum Sci. Technol.* **10**, 025034 (2025); [arXiv:2411.00468](#).
- [12] J. Yu *et al.* (incl. M. Motta, A. Mezzacapo), “Quantum-centric algorithm for sample-based Krylov diagonalization,” [arXiv:2501.09702](#) (2025).
- [13] W. Kirby, M. Motta, A. Mezzacapo, “Exact and efficient Lanczos method on a quantum computer,” *Quantum* **7**, 1018 (2023); [arXiv:2208.00567](#).
- [14] N. Yoshioka *et al.* (incl. A. Mezzacapo), “Diagonalization of large many-body Hamiltonians on a quantum processor,” *Nat. Commun.* **16**, 5014 (2025); [arXiv:2407.14431](#).
- [15] T. Shirakawa *et al.*, “Closed-loop calculations of electronic structure on a quantum processor and a classical supercomputer at full scale,” [arXiv:2511.00224](#) (2025).
- [16] D. Danilov, J. Robledo-Moreno, K. J. Sung, M. Motta, A. Shee, “Enhancing the accuracy and efficiency of sample-based quantum diagonalization with phaseless auxiliary-field quantum Monte Carlo,” [arXiv:2503.05967](#) (2025).
- [17] A. Shajan *et al.* (incl. M. Motta, A. Mezzacapo), “Towards quantum-centric simulations of extended molecules: sample-based quantum diagonalization enhanced with density matrix embedding theory,” [arXiv:2411.09861](#) (2024).
- [18] T. Ohgoe, H. Iwakiri, K. Ichikawa, S. Koh, M. Kohda, “Quantum computation of a quasiparticle band structure with the quantum-selected configuration interaction,” *J. Phys. Soc. Jpn.* **94**, 114002 (2025); [arXiv:2504.00309](#).
- [19] K. Nogaki, S. Backes, T. Shirakawa, S. Yunoki, R. Arita, “Symmetry-adapted sample-based quantum diagonalization: application to lattice model,” [arXiv:2505.00914](#) (2025).
- [20] S. Lee, J. Lee, H. Zhai *et al.* (incl. D. R. Reichman, L. Lin, G. K.-L. Chan), “Is there evidence for exponential quantum advantage in quantum chemistry?,” *Nat. Commun.* **14**, 1952 (2023); [arXiv:2208.02199](#).
- [21] Z. Hassman, O. Reardon-Smith, G. S. Ravi, F. T. Chong, K. J. Sung, “ExtraFerm: an extended matchgate simulator,” [arXiv:2511.12416](#) (2025).
- [22] H. Belagali *et al.* (incl. R. LaRose), “Efficient classical simulation of large-scale unitary cluster Jastrow circuits,” [arXiv:2607.21337](#) (2026).
- [23] J.-Y. Chang *et al.*, “An iterative dual-channel neural quantum state algorithm for selected configuration interaction,” [arXiv:2606.26760](#) (2026).
- [24] S. Thompson, D. Gunlycke, “Auto-regressive neural quantum state sampling for selected configuration interaction,” [arXiv:2603.24728](#) (2026).
- [25] P. Reinholdt, K. M. Ziems, E. R. Kjellgren, S. Coriani, S. P. A. Sauer, J. Kongsted, “Critical limitations in quantum-selected configuration interaction methods,” *J. Chem. Theory Comput.* **21**, 6811 (2025); [arXiv:2501.07231](#).
- [26] C. Gaberle, M. S. Jattana, “A critical assessment of the sample-based quantum diagonalization for Heisenberg and Hubbard models,” [arXiv:2605.02494](#) (2026).
- [27] N. Bonilla Vargas, “Machine learning for sample-based quantum diagonalization: generative configuration recovery and the classical-simulability frontier,” [arXiv:2608.05314](#) (2026).

- [28] Y. Kim, A. Eddins, S. Anand *et al.* (IBM Quantum), “Evidence for the utility of quantum computing before fault tolerance,” *Nature* **618**, 500 (2023); DOI [10.1038/s41586-023-06096-3](https://doi.org/10.1038/s41586-023-06096-3).
- [29] J. Tindall, M. Fishman, E. M. Stoudenmire, D. Sels, “Efficient tensor network simulation of IBM’s Eagle kicked Ising experiment,” *PRX Quantum* **5**, 010308 (2024); [arXiv:2306.14887](https://arxiv.org/abs/2306.14887).
- [30] T. Begušić, J. Gray, G. K.-L. Chan, “Fast and converged classical simulations of evidence for the utility of quantum computing before fault tolerance,” *Sci. Adv.* **10**, eadk4321 (2024); [arXiv:2308.05077](https://arxiv.org/abs/2308.05077).
- [31] R. Babbush, J. R. McClean, M. Newman, C. Gidney, S. Boixo, H. Neven, “Focus beyond quadratic speedups for error-corrected quantum advantage,” *PRX Quantum* **2**, 010103 (2021); [arXiv:2011.04149](https://arxiv.org/abs/2011.04149).
- [32] G. Mazzola, “Quantum computing for chemistry and physics applications from a Monte Carlo perspective,” *J. Chem. Phys.* **160**, 010901 (2024); [arXiv:2308.07964](https://arxiv.org/abs/2308.07964).
- [33] Google Quantum AI, “Observation of constructive interference at the edge of quantum ergodicity,” *Nature* **646**, 825 (2025); [arXiv:2506.10191](https://arxiv.org/abs/2506.10191).
- [34] F. Arute *et al.* (Google AI Quantum), “Quantum supremacy using a programmable superconducting processor,” *Nature* **574**, 505 (2019); DOI [10.1038/s41586-019-1666-5](https://doi.org/10.1038/s41586-019-1666-5).
- [35] Google Quantum AI, “Programmable digital quantum simulation of 2D Fermi-Hubbard dynamics using 72 superconducting qubits,” [arXiv:2510.26845](https://arxiv.org/abs/2510.26845) (2025).
- [36] Quantinuum, “Fermionic dynamics on a trapped-ion quantum computer beyond exact classical simulation,” [arXiv:2510.26300](https://arxiv.org/abs/2510.26300) (2025).
- [37] Qedma, IBM, RIKEN *et al.*, “Resolving structure in prethermal Floquet dynamics with precision quantum computation,” [arXiv:2607.24937](https://arxiv.org/abs/2607.24937) (2026).
- [38] Algorithmiq, IBM *et al.*, “Observable estimation in the absence of classical verification,” [arXiv:2607.25998](https://arxiv.org/abs/2607.25998) (2026).
- [39] E. Granet *et al.* (Quantinuum), “Dynamical structure factor with a pumping approach on a trapped-ion quantum computer,” [arXiv:2607.07138](https://arxiv.org/abs/2607.07138) (2026).
- [40] Y. Shen *et al.*, “Real-time Krylov theory for quantum computing algorithms,” *Quantum* **7**, 1066 (2023); [arXiv:2208.01063](https://arxiv.org/abs/2208.01063).
- [41] J. R. McClean, M. E. Kimchi-Schwartz, J. Carter, W. A. de Jong, “Hybrid quantum-classical hierarchy for mitigation of decoherence and determination of excited states,” *Phys. Rev. A* **95**, 042308 (2017); [arXiv:1603.05681](https://arxiv.org/abs/1603.05681).
- [42] C. L. Cortes, S. K. Gray, “Quantum Krylov subspace algorithms for ground- and excited-state energy estimation,” *Phys. Rev. A* **105**, 022417 (2022); [arXiv:2109.06868](https://arxiv.org/abs/2109.06868).
- [43] R. M. Parrish, P. L. McMahon, “Quantum filter diagonalization: quantum eigendecomposition without full quantum phase estimation,” [arXiv:1909.08925](https://arxiv.org/abs/1909.08925) (2019).
- [44] E. N. Epperly, L. Lin, Y. Nakatsukasa, “A theory of quantum subspace diagonalization,” *SIAM J. Matrix Anal. Appl.* **43**, 1263 (2022); [arXiv:2110.07492](https://arxiv.org/abs/2110.07492).
- [45] I. Rungger *et al.*, “Dynamical mean field theory algorithm and experiment on quantum computers,” [arXiv:1910.04735](https://arxiv.org/abs/1910.04735) (2020).
- [46] C. Umeano, F. Jamet, L. P. Lindoy, I. Rungger, O. Kyriienko, “Quantum subspace expansion approach for simulating dynamical response functions of Kitaev spin liquids,” *Phys. Rev. Materials* **9**, 034401 (2025); [arXiv:2407.04205](https://arxiv.org/abs/2407.04205).

- [47] B. Bauer, D. Wecker, A. J. Millis, M. B. Hastings, M. Troyer, “Hybrid quantum-classical approach to correlated materials,” *Phys. Rev. X* **6**, 031045 (2016); [arXiv:1510.03859](#).
- [48] S. Endo, I. Kurata, Y. O. Nakagawa, “Calculation of the Green’s function on near-term quantum computers,” *Phys. Rev. Research* **2**, 033281 (2020); [arXiv:1909.12250](#).
- [49] G. Greene-Diniz, D. Z. Manrique, K. Yamamoto *et al.* (Quantinuum), “Quantum computed Green’s functions using a cumulant expansion of the Lanczos method,” *Quantum* **8**, 1383 (2024); [arXiv:2309.09685](#).
- [50] T. A. Bespalova, K. Delić, G. Pupillo, F. Tacchino, I. Tavernelli, “Simulating the Fermi-Hubbard model with long-range hopping on a quantum computer,” *Phys. Rev. A* **111**, 052619 (2025); [arXiv:2410.07789](#).
- [51] A. Santini, S. Barison, F. Vicentini, “Classically prepared, quantumly evolved: a hybrid algorithm for molecular spectra,” [arXiv:2510.24911](#) (2025).
- [52] W. Du *et al.* (incl. J. P. Vary), “Quantum-classical framework for many-fermion response and structure,” [arXiv:2602.08357](#) (2026).
- [53] G. H. Golub, G. Meurant, *Matrices, Moments and Quadrature with Applications* (Princeton Univ. Press, 2010).
- [54] R. Haydock, V. Heine, M. J. Kelly, “Electronic structure based on the local atomic environment for tight-binding bands,” *J. Phys. C* **5**, 2845 (1972); R. Haydock, “The recursive solution of the Schrödinger equation,” *Solid State Phys.* **35**, 215 (1980).
- [55] S. Piccinelli, A. Baiardi, S. Barison, M. Rossmannek, A. Carrera Vazquez, F. Tacchino *et al.* (incl. A. Alavi, M. Motta), “Quantum chemistry with provable convergence via randomized sample-based Krylov quantum diagonalization,” [arXiv:2508.02578](#) (2025).
- [56] B. Huron, J.-P. Malrieu, P. Rancurel, “Iterative perturbation calculations of ground and excited state energies from multiconfigurational zeroth-order wavefunctions,” *J. Chem. Phys.* **58**, 5745 (1973).
- [57] Y. Garniron *et al.*, “Quantum Package 2.0: an open-source determinant-driven suite of programs,” *J. Chem. Theory Comput.* **15**, 3591 (2019); [arXiv:1902.08154](#).
- [58] G. L. Squires, *Introduction to the Theory of Thermal Neutron Scattering*, 3rd ed. (Cambridge Univ. Press, 2012).
- [59] L. J. P. Ament, M. van Veenendaal, T. P. Devereaux, J. P. Hill, J. van den Brink, “Resonant inelastic x-ray scattering studies of elementary excitations,” *Rev. Mod. Phys.* **83**, 705 (2011).
- [60] M. Troyer, U.-J. Wiese, “Computational complexity and fundamental limitations to fermionic quantum Monte Carlo simulations,” *Phys. Rev. Lett.* **94**, 170201 (2005).
- [61] P. Calabrese, J. Cardy, “Evolution of entanglement entropy in one-dimensional systems,” *J. Stat. Mech.* P04010 (2005).
- [62] N. Schuch, F. Verstraete, “Computational complexity of interacting electrons and fundamental limitations of density functional theory,” *Nat. Phys.* **5**, 732 (2009); [arXiv:0712.0483](#).
- [63] N. Vaquero-Sabater, A. Carreras, L. Broers, T. Shirakawa, S. Yunoki, D. Casanova, “Noise and configuration recovery impact on quantum-selected configuration interaction,” [arXiv:2605.23697](#) (2026).
- [64] L. G. Valiant, “Quantum circuits that can be simulated classically in polynomial time,” *SIAM J. Comput.* **31**, 1229 (2002).

- [65] B. M. Terhal, D. P. DiVincenzo, “Classical simulation of noninteracting-fermion quantum circuits,” *Phys. Rev. A* **65**, 032325 (2002); [quant-ph/0108010](#).
- [66] S. Bravyi, “Lagrangian representation for fermionic linear optics,” *Quantum Inf. Comput.* **5**, 216 (2005); [quant-ph/0404180](#).
- [67] R. Jozsa, A. Miyake, “Matchgates and classical simulation of quantum circuits,” *Proc. R. Soc. A* **464**, 3089 (2008); [arXiv:0804.4050](#).
- [68] M. Hebenstreit, R. Jozsa, B. Kraus, S. Strelchuk, M. Yoganathan, “All pure fermionic non-Gaussian states are magic states for matchgate computations,” *Phys. Rev. Lett.* **123**, 080503 (2019); [arXiv:1905.08584](#).
- [69] S. Bravyi, A. Kitaev, “Universal quantum computation with ideal Clifford gates and noisy ancillas,” *Phys. Rev. A* **71**, 022316 (2005); [quant-ph/0403025](#).
- [70] L. Leone, S. F. E. Oliviero, A. Hama, “Stabilizer Rényi entropy,” *Phys. Rev. Lett.* **128**, 050402 (2022); [arXiv:2106.12587](#).
- [71] P. Sierant, P. Stornati, X. Turkeshi, “Fermionic magic resources of quantum many-body systems,” *PRX Quantum* **7**, 010302 (2026); [arXiv:2506.00116](#).
- [72] R. Świątek, M. Kliczkowski, L. Vidmar, M. Rigol, “One-body purity, non-Gaussianity, and entanglement in interacting integrable models,” [arXiv:2607.01326](#) (2026).
- [73] M. Dias, R. Koenig, “Classical simulation of non-Gaussian fermionic circuits,” *Quantum* **8**, 1350 (2024); [arXiv:2307.12912](#).
- [74] O. Reardon-Smith, M. Oszmaniec, K. Korzekwa, “Improved simulation of quantum circuits dominated by free fermionic operations,” *Quantum* **8**, 1549 (2024); [arXiv:2307.12702](#).
- [75] J. Cudby, S. Strelchuk, “Gaussian decomposition of magic states for matchgate computations,” [arXiv:2307.12654](#) (2023).
- [76] P. S. Tarabunga, B. Jobst, R. Morral-Yepes, M. Langer, B. Kraus, F. Pollmann, S.-H. Lin, “Computable measures of fermionic non-Gaussianity from the covariance matrix,” [arXiv:2607.02242](#) (2026).
- [77] C. Oh, M. Oszmaniec, O. Reardon-Smith, Z. Zimborás, “Classical simulation of free-fermionic dynamics and quantum chemistry with magic input,” [arXiv:2604.26813](#) (2026).
- [78] M. Collura, J. De Nardis, V. Alba, G. Lami, “The non-stabilizerness of fermionic Gaussian states,” *Quantum* **10**, 2036 (2026); [arXiv:2412.05367](#).
- [79] L. Bittel, A. A. Mele, J. Eisert, L. Leone, “Optimal trace-distance bounds for free-fermionic states: testing and improved tomography,” *PRX Quantum* **6**, 030341 (2025); [arXiv:2409.17953](#).
- [80] A. A. Mele, Y. Herasymenko, “Efficient learning of quantum states prepared with few fermionic non-Gaussian gates,” *PRX Quantum* **6**, 010319 (2025); [arXiv:2402.18665](#).
- [81] M. Sarkis, A. Tkatchenko, “Are Molecules Magical? Non-Stabilizerness in Molecular Bonding,” [arXiv:2504.06673](#) (2025).
- [82] A. F. Kemper *et al.*, “Efficient computation of real-time correlators using Pauli propagation,” [arXiv:2607.24924](#) (2026).
- [83] S. Paeckel *et al.*, “Time-evolution methods for matrix-product states,” *Ann. Phys.* **411**, 167998 (2019); [arXiv:1901.05824](#).

- [84] M. J. Bremner, A. Montanaro, D. J. Shepherd, “Average-case complexity versus approximate simulation of commuting quantum computations,” *Phys. Rev. Lett.* **117**, 080501 (2016); [arXiv:1504.07999](#).
- [85] A. Hafid, H. Iwakiri, K. Tsubouchi, N. Yoshioka, M. Kohda, “Hardness of classically sampling quantum chemistry circuits,” [arXiv:2504.12893](#) (2025).
- [86] M. Hinsche *et al.*, “A single T -gate makes distribution learning hard,” *Phys. Rev. Lett.* **130**, 240602 (2023); [arXiv:2207.03140](#).
- [87] S. R. White, “Density matrix formulation for quantum renormalization groups,” *Phys. Rev. Lett.* **69**, 2863 (1992); DOI [10.1103/PhysRevLett.69.2863](#).
- [88] T. D. Kühner, S. R. White, “Dynamical correlation functions using the density matrix renormalization group,” *Phys. Rev. B* **60**, 335 (1999); DOI [10.1103/PhysRevB.60.335](#).
- [89] E. Jeckelmann, “Dynamical density-matrix renormalization-group method,” *Phys. Rev. B* **66**, 045114 (2002); [cond-mat/0203500](#).
- [90] A. Holzner, A. Weichselbaum, I. P. McCulloch, U. Schollwöck, J. von Delft, “Chebyshev matrix product state approach for spectral functions,” *Phys. Rev. B* **83**, 195115 (2011); DOI [10.1103/PhysRevB.83.195115](#).
- [91] A. Nocera, G. Alvarez, “Spectral functions with the density matrix renormalization group: Krylov-space correction vectors,” *Phys. Rev. E* **94**, 053308 (2016); [arXiv:1607.03538](#).
- [92] S. R. White, A. E. Feiguin, “Real-time evolution using the density matrix renormalization group,” *Phys. Rev. Lett.* **93**, 076401 (2004); DOI [10.1103/PhysRevLett.93.076401](#).
- [93] A. Weiße, G. Wellein, A. Alvermann, H. Fehske, “The kernel polynomial method,” *Rev. Mod. Phys.* **78**, 275 (2006); DOI [10.1103/RevModPhys.78.275](#).
- [94] A. A. Holmes, N. M. Tubman, C. J. Umrigar, “Heat-bath configuration interaction: an efficient selected configuration interaction algorithm inspired by heat-bath sampling,” *J. Chem. Theory Comput.* **12**, 3674 (2016); [arXiv:1606.07453](#).
- [95] S. Sharma, A. A. Holmes, G. Jeanmairet, A. Alavi, C. J. Umrigar, “Semistochastic heat-bath configuration interaction method,” *J. Chem. Theory Comput.* **13**, 1595 (2017); [arXiv:1610.06660](#).
- [96] N. M. Tubman, J. Lee, T. Y. Takeshita, M. Head-Gordon, K. B. Whaley, “A deterministic alternative to the full configuration interaction quantum Monte Carlo method,” *J. Chem. Phys.* **145**, 044112 (2016); [arXiv:1603.02686](#).
- [97] G. K.-L. Chan, M. Head-Gordon, “Highly correlated calculations with a polynomial cost algorithm: a study of the density matrix renormalization group,” *J. Chem. Phys.* **116**, 4462 (2002); DOI [10.1063/1.1449459](#).
- [98] H. Zhai, H. R. Larsson, S. Lee *et al.* (G. K.-L. Chan), “Block2: a comprehensive open source framework to develop and apply state-of-the-art DMRG algorithms in electronic structure and beyond,” *J. Chem. Phys.* **159**, 234801 (2023); [arXiv:2310.03920](#).
- [99] M. Motta, S. Zhang, “Ab initio computations of molecular systems by the auxiliary-field quantum Monte Carlo method,” *WIREs Comput. Mol. Sci.* **8**, e1364 (2018); [arXiv:1711.02242](#).
- [100] A. Shee, Z. Huang, M. Head-Gordon, K. B. Whaley, “Real-time propagation of adaptive sampling selected configuration interaction wave function,” [arXiv:2411.07615](#) (2024).
- [101] A. Georges, G. Kotliar, W. Krauth, M. J. Rozenberg, “Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions,” *Rev. Mod. Phys.* **68**, 13 (1996); DOI [10.1103/RevModPhys.68.13](#).

- [102] A. Damascelli, Z. Hussain, Z.-X. Shen, “Angle-resolved photoemission studies of the cuprate superconductors,” *Rev. Mod. Phys.* **75**, 473 (2003); DOI [10.1103/RevModPhys.75.473](https://doi.org/10.1103/RevModPhys.75.473).
- [103] H.-Y. Huang *et al.*, “Quantum advantage in learning from experiments,” *Science* **376**, 1182 (2022); [arXiv:2112.00778](https://arxiv.org/abs/2112.00778).
- [104] E. Bengio, M. Jain, M. Korablyov, D. Precup, Y. Bengio, “Flow network based generative models for non-iterative diverse candidate generation,” *NeurIPS* **34** (2021); [arXiv:2106.04399](https://arxiv.org/abs/2106.04399).
- [105] C. Patra, D. Mondal *et al.* (R. Maitra), “Physics-informed generative machine learning for accelerated quantum-centric supercomputing,” [arXiv:2512.06858](https://arxiv.org/abs/2512.06858) (2025).
- [106] I. L. Huidobro-Meezs, J. Dai, G. Rabusseau, R. A. Vargas-Hernández, “GFlowNets for Hamiltonian decomposition in groups of compatible operators,” *Machine Learning and the Physical Sciences Workshop, NeurIPS 2024*; [arXiv:2410.16041](https://arxiv.org/abs/2410.16041) (2024).
- [107] I. L. Huidobro-Meezs, J. Dai, R. A. Vargas-Hernández, “Discrete flow-based generative models for measurement optimization in quantum computing,” [arXiv:2509.15486](https://arxiv.org/abs/2509.15486) (2025).
- [108] R. A. Vargas-Hernández, Y. Guan, D. H. Zhang, R. V. Krems, “Bayesian optimization for the inverse scattering problem in quantum reaction dynamics,” *New J. Phys.* **21**, 022001 (2019); [arXiv:1711.06376](https://arxiv.org/abs/1711.06376).
- [109] W. J. Huggins, B. A. O’Gorman, N. C. Rubin, D. R. Reichman, R. Babbush, J. Lee, “Unbiasing fermionic quantum Monte Carlo with a quantum computer,” *Nature* **603**, 416 (2022); [arXiv:2106.16235](https://arxiv.org/abs/2106.16235).
- [110] D. Aharonov *et al.* (N. H. Lindner), “Reliable high-accuracy error mitigation for utility-scale quantum circuits,” [arXiv:2508.10997](https://arxiv.org/abs/2508.10997) (2025).
- [111] M. Reiher, N. Wiebe, K. M. Svore, D. Wecker, M. Troyer, “Elucidating reaction mechanisms on quantum computers,” *Proc. Natl. Acad. Sci. USA* **114**, 7555 (2017); [arXiv:1605.03590](https://arxiv.org/abs/1605.03590).
- [112] K. Takatsuka, T. Fueno, K. Yamaguchi, “Distribution of odd electrons in ground-state molecules,” *Theor. Chim. Acta* **48**, 175 (1978).
- [113] V. N. Staroverov, E. R. Davidson, “Distribution of effectively unpaired electrons,” *Chem. Phys. Lett.* **330**, 161 (2000).
- [114] M. Head-Gordon, “Characterizing unpaired electrons from the one-particle density matrix,” *Chem. Phys. Lett.* **372**, 508 (2003).
- [115] N. Gigena, M. Di Tullio, R. Rossignoli, “One-body entanglement as a quantum resource in fermionic systems,” *Phys. Rev. A* **102**, 042410 (2020); [arXiv:2001.03570](https://arxiv.org/abs/2001.03570).
- [116] L. Coffman, G. Smith, X. Gao, “Measuring non-Gaussian magic in fermions: convolution, entropy, and the violation of Wick’s theorem,” [arXiv:2501.06179](https://arxiv.org/abs/2501.06179) (2025).
- [117] V. de Silva, L.-H. Lim, “Tensor rank and the ill-posedness of the best low-rank approximation problem,” *SIAM J. Matrix Anal. Appl.* **30**, 1084 (2008); [arXiv:math/0607647](https://arxiv.org/abs/math/0607647).
- [118] H. Zhang, M. Otten, “From random determinants to the ground state,” [arXiv:2511.14734](https://arxiv.org/abs/2511.14734) (2025).
- [119] L. A. Wray, C.-J. Lin, V. Su, H. Gharibyan, “Convergence of sample-based quantum diagonalization on a variable-length cuprate chain,” [arXiv:2512.04962](https://arxiv.org/abs/2512.04962) (2025).
- [120] B. Seibert, S. Alterman, Q. Wang, F. Qian, A. Miyake, P. J. Love, “Correlation is magic in electronic structure Hamiltonians,” [arXiv:2606.31799](https://arxiv.org/abs/2606.31799) (2026).

- [121] R. Walkup *et al.* (incl. J. Robledo-Moreno, W. Kirby, A. Mezzacapo, A. Córcoles), “Scaling sample-based quantum diagonalization on GPU-accelerated systems using OpenMP offload,” [arXiv:2601.16169](#) (2026).
- [122] J. Doi, T. Shirakawa, Y. Kawashima, S. Yunoki, H. Horii, “GPU-accelerated selected basis diagonalization with Thrust for SQD-based algorithms,” [arXiv:2601.16637](#) (2026).
- [123] E. R. Bennewitz, F. Hopfmueller, B. Kulchytskyy, J. Carrasquilla, P. Ronagh, “Neural error mitigation of near-term quantum simulations,” *Nat. Mach. Intell.* **4**, 618 (2022); [arXiv:2105.08086](#).
- [124] M. J. Solanki, L. Ding, M. Reiher, “Neural quantum states based on selected configurations,” [arXiv:2602.12993](#) (2026).
- [125] S. Zeni, G. Varutti, J. Nespolo, D. Trypogeorgos, “Enhancing quantum-classical configuration interaction methods using a neural-network classifier,” [arXiv:2606.24332](#) (2026).
- [126] H.-Y. Huang, M. Broughton, N. Eassa, H. Neven, R. Babbush, J. R. McClean, “Generative quantum advantage for classical and quantum problems,” [arXiv:2509.09033](#) (2025).
- [127] A. Huang, W. Maxwell, V. Belis, E. Peters, J. Pye, S. Jahangiri, J. Bowles, “Spectral Born machines: classically trainable quantum generative models for discrete data,” [arXiv:2607.06675](#) (2026).
- [128] L. Zhao *et al.*, “Quantum-classical auxiliary-field quantum Monte Carlo with matchgate shadows on trapped-ion quantum computers,” [arXiv:2506.22408](#) (2025).
- [129] M. Oszmaniec, N. Dangniam, M. E. S. Morales, Z. Zimborás, “Fermion sampling: a robust quantum computational advantage scheme using fermionic linear optics and magic input states,” *PRX Quantum* **3**, 020328 (2022); [arXiv:2012.15825](#).
- [130] B. Gauthier, P. Rosenberg, A. Foley, M. Charlebois, “An occupation-number quantum subspace expansion approach to compute the single-particle Green function: an opportunity for noise filtering,” *Phys. Rev. A* **110**, 032624 (2024); [arXiv:2312.13497](#).
- [131] U. Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Ann. Phys.* **326**, 96 (2011); [arXiv:1008.3477](#).
- [132] M. Frau, P. S. Tarabunga, M. Collura, M. Dalmonte, E. Tirrito, “Non-stabilizerness versus entanglement in matrix product states,” *Phys. Rev. B* **110**, 045101 (2024); [arXiv:2404.18768](#).
- [133] E. H. Lieb, F. Y. Wu, “Absence of Mott transition in an exact solution of the short-range, one-band model in one dimension,” *Phys. Rev. Lett.* **20**, 1445 (1968); DOI [10.1103/PhysRevLett.20.1445](#).
- [134] J. des Cloizeaux, J. J. Pearson, “Spin-wave spectrum of the antiferromagnetic linear chain,” *Phys. Rev.* **128**, 2131 (1962); DOI [10.1103/PhysRev.128.2131](#).
- [135] G. Müller, H. Thomas, H. Beck, J. C. Bonner, “Quantum spin dynamics of the antiferromagnetic linear chain in zero and nonzero magnetic field,” *Phys. Rev. B* **24**, 1429 (1981); DOI [10.1103/PhysRevB.24.1429](#).